



Betting against betting against beta[☆]

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ABSTRACT

Frazzini and Pedersen's (2014) Betting Against Beta (BAB) factor is based on the same basic idea as Blacks' (1972) beta-arbitrage, but its astonishing performance has generated academic interest and made it highly influential with practitioners. This performance is driven by non-standard procedures used in its construction that effectively, but non-transparently, equal weight stock returns. For each dollar invested in BAB, the strategy commits on average \$1.05 to stocks in the bottom 1% of total market capitalization. BAB earns positive returns after accounting for transaction costs, but earns these by tilting toward profitability and investment. Predictable biases resulting from Frazzini and Pedersen's non-standard beta estimation procedure drive results presented as evidence supporting BAB's underlying theory.

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1. Introduction

Frazzini and Pedersen (2014) Betting Against Beta (BAB, 2014) is an unmitigated academic success. It is, at the time of this writing, the third most downloaded article from the *Journal of Financial Economics* over the last 90 days, and its field-weighted citation impact suggests it has been cited almost 27 times more often than the average paper published in similar journals. Its impact on practice has been even greater. It is one of the most influential articles on “defensive equity,” a class of strategies that has seen mas-

sive capital inflows and is now a major investment category for institutional investors.

This success is particularly remarkable because BAB is based on a fairly simple idea that predates its publication by more than 40 years. Black (1972) first suggested trading to exploit an empirical failure of the Capital Asset Pricing Model (CAPM): the fact that the observed Capital Market Line is “too flat.” Low beta stocks have earned higher average returns than predicted by the CAPM, while high beta stocks have underperformed the model's predictions, so strategies that overweight low beta stocks and underweight high beta stocks have earned positive CAPM alphas.

Despite being widely read, and based on a fairly simple idea, BAB is not well understood. This is because the authors use three unconventional procedures to construct their factor. All three departures from standard factor construction contribute to the paper's strong empirical results. None is important for understanding the underlying economics, and each obscures the mechanisms driving reported effects.

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Two of these non-standard procedures drive BAB's astonishing "paper" performance, which cannot be achieved in practice, while the other drives results [Frazzini and Pedersen \(2014\)](#) present as evidence supporting their theory. The two responsible for driving performance can be summarized as follows:

- **Non-standard procedure #1: rank-weighted portfolio construction.** Instead of simply sorting stocks into value-weighted portfolios when constructing the beta portfolios underlying BAB, [Frazzini and Pedersen \(2014\)](#) use a "rank-weighting" procedure that assigns each stock to either the "high" portfolio or the "low" portfolio with a weight proportional to the cross-sectional deviation of the stock's estimated beta rank from the median rank.
- **Non-standard procedure #2: hedging by leveraging.** Instead of hedging the low-minus-high beta strategy underlying BAB by buying the market in proportion to the underlying strategy's observed short market tilt, [Frazzini and Pedersen \(2014\)](#) attempt to achieve market-neutrality by leveraging the low beta portfolio and deleveraging the high beta portfolio using these portfolios' predicted betas, with the intention that the scaled portfolios' betas are each equal to one and thus net to zero in the long/short strategy.

[Frazzini and Pedersen \(2014\)](#)'s first non-standard procedure, rank-weighting, drives BAB's performance not by what it does, i.e., put more weight on stocks with extreme betas, but by what it does *not* do, i.e., weight stocks in proportion to their market capitalizations, as is standard in asset pricing. The procedure creates portfolios that are almost indistinguishable from simple, equal-weighted portfolios. Their second non-standard procedure, hedging with leverage, uses these same portfolios to hedge the low-minus-high beta strategy underlying BAB. That is, the rank-weighting procedure is a backdoor to equal-weighting the underlying beta portfolios, and the leveraging procedure is a backdoor to using equal-weighted portfolios for hedging.

BAB achieves its high Sharpe ratio, and large, highly significant alpha relative to the common factor models, by hugely overweighting micro- and nano-cap stocks. For each dollar invested in BAB, the strategy commits on average \$1.05 to stocks in the bottom 1% of total market capitalization. These stocks have limited capacity and are expensive to trade. As a result, while BAB's "paper" performance is impressive, it is not something an investor can actually realize.

Accounting for transaction costs reduces BAB's profitability by almost 60%. While it still earns significant positive returns, these returns are explained by its tilts toward profitability and investment; it does not have a significant net alpha relative to the [Fama and French \(2015\)](#) five-factor model.

The third non-standard procedure [Frazzini and Pedersen \(2014\)](#) use when constructing BAB, a novel method for estimating betas, mechanically generates results they present as empirical evidence supporting their paper's underlying theory. One of the paper's main theoretical predictions is "beta compression," specifically that the market "betas of securities in the cross section are compressed

toward one when funding liquidity risk is high" (p. 2). To support this prediction, they present evidence that the cross-sectional dispersion in betas is negatively correlated with Treasury-Eurodollar-spread (TED-spread) volatility, which they use as a proxy for funding liquidity risk. This result actually reflects biases resulting from their non-standard beta estimation procedure.

[Frazzini and Pedersen \(2014\)](#) non-standard beta estimation procedure works as follows:

- **Non-standard procedure #3: novel beta estimation technique.** Instead of estimating betas as slope coefficients from CAPM regressions, [Frazzini and Pedersen \(2014\)](#) measure betas by combining market correlations estimated using five years of overlapping three-day returns with volatilities estimated using one year of daily data.

[Frazzini and Pedersen \(2014\)](#) justify this procedure by stating that they estimate betas using "one-year rolling standard deviation for volatilities and a five-year horizon for the correlation to account for the fact that correlations appear to move more slowly than volatilities" (p. 8). While this justification may sound appealing, and the method has seen significant adoption in the literature that follows BAB, the [Frazzini and Pedersen \(2014\)](#) beta estimates (FP-betas) are strongly biased in predictable ways. This can be seen using the following identity for the i th asset's FP-beta,

$$\beta_{FP}^i = \left(\frac{\rho_5^i}{\rho_1^i} \right) \beta_1^i, \quad (1)$$

where ρ^i is used to denote the asset's correlation with the market, β^i is its beta estimated from a CAPM regression, and numeric subscripts denote estimation windows measured in years.

The preceding equation suggests that FP-betas are biased in ways that can be predicted using market volatility. Individual stocks tend to be more correlated when markets are volatile, because high correlations drive market volatility. Empirically a one standard deviation increase in the market volatility realized over the preceding year is associated with a highly significant half standard deviation increase in the average correlation realized between individual stocks and the market. Stock correlations estimated at longer horizons tend to be less variable, so individual FP-beta estimates tend to be biased down when markets are unsettled and the average ρ_1^i is high, and biased up when markets are calm and the average ρ_1^i is low.

Direct evidence for these facts can be seen by looking at the market's FP-beta, calculated as the weighted average FP-beta of the individual stocks in the market portfolio, the method [Frazzini and Pedersen \(2014\)](#) use for calculating portfolio betas. The value-weighted average beta of all the stocks in the market must always be exactly one, so the extent to which the value-weighted average of any beta estimator differs from one quantifies the value-weighted average bias in the estimator. The FP-betas are quite biased. The value-weighted average FP-beta has a time-series mean of 1.05, and a standard deviation of 0.09. It frequently exceeds 1.2, hitting a high of 1.44 in November 1964. Market volatility explains 45% of its time-series variation.

This highly predictable bias in the FP-beta estimates drives Frazzini and Pedersen (2014) results on beta compression. The bias impacts the average stock, mechanically compressing the distribution of FP-beta estimates whenever market volatility is high. Any variable correlated with market volatility consequently predicts “beta compression,” at least when betas are estimated using Frazzini and Pedersen (2014) novel procedure. The Frazzini and Pedersen (2014) results on beta compression, which use FP-betas, consequently suggest that TED-spread volatility is correlated with market volatility (it is), but is not strong evidence in favor of the authors’ model. Tests performed using conventional market beta estimates instead of the biased FP-betas provide no support for the paper’s theory.

The biased FP-betas are also used in Frazzini and Pedersen (2014) hedging procedure, with the result that BAB is not market-neutral, contrary to Frazzini and Pedersen (2014) intent. Market volatility predicts the bias in the FP-beta estimates of the underlying beta portfolios, but not the realized future betas of these underlying portfolios. Market volatility consequently also predicts the severity of BAB’s mis-hedging, and thus its conditional beta. This fact makes the results that Frazzini and Pedersen (2014) present on BAB’s conditional market tilts difficult to interpret. They offer the results as support for the paper’s underlying theory, but because of their non-standard construction methods any variable correlated with market volatility mechanically predicts BAB’s mis-hedging, and thus its conditional beta.

These facts raise a question: how can a paper be so influential, and so well read, yet so poorly understood? Our paper provides an answer, by highlighting the role that each of BAB’s non-standard, non-transparent procedures plays in generating its strong results. While our paper’s specific focus is BAB, its intended message is general: results dependent on non-standard methods should be evaluated cautiously.

2. Alternative procedures driving BAB’s performance

This section quantifies the impact Frazzini and Pedersen (2014) non-standard portfolio construction procedure and non-standard hedging procedure have on BAB’s performance.¹ It does so by comparing the performance of BAB to similarly constructed “almost BAB” strategies, which differ from BAB only by using standard methods for either portfolio construction or hedging.

These comparisons obviously require the performance of BAB, the yardstick against which we measure the “almost BAB” variations. This involves a choice, as there is not a single “canonical” BAB strategy. BAB’s returns are maintained by AQR Capital Management. While Frazzini and Pedersen (2014) present some results for other asset classes, AQR only updates the equity factors, and we similarly limit our focus. AQR makes available the “original paper dataset,” which ends after March 2012, and a “BAB equity factor,” which is “an updated and extended version

of the paper data.”² Unfortunately these two factors differ. The monthly correlation between the two US equity factors is 97.5% in the post-1967 sample, but the original series has a significantly higher Sharpe ratio than the currently maintained series.³ When we replicate BAB using the construction methods described by Frazzini and Pedersen (2014), our factor has a monthly correlation of 98.0% with Frazzini and Pedersen (2014) original series, and realizes the same 1.01 Sharpe ratio over the January 1968 to March 2012 sample period.⁴

Overall, our BAB replication is quite similar to the original paper series, and is directly comparable to the “almost BAB” strategies we consider next, because it is constructed using exactly the same data. It is also available beyond the original series’ March 2012 end date. We consequently use our replication of BAB as the yardstick against which we evaluate the “almost BAB” strategies.

2.1. Rank-weighting versus standard portfolio sorts

When Frazzini and Pedersen (2014) construct the beta portfolios underlying BAB, the weight on each stock is proportional to the difference between the stock’s cross-sectional characteristic rank, r_i , and the median rank, r_{median} ,

$$w_i = \begin{cases} \frac{(r_i - r_{\text{median}})}{\sum_{\{j|r_j < r_{\text{median}}\}} (r_j - r_{\text{median}})} & \text{if } r_i < r_{\text{median}} \text{ (low beta portfolio),} \\ \frac{(r_i - r_{\text{median}})}{\sum_{\{j|r_j > r_{\text{median}}\}} (r_j - r_{\text{median}})} & \text{if } r_i > r_{\text{median}} \text{ (high beta portfolio).} \end{cases}$$

Note that firm size is not considered when constructing rank-weighted portfolios.⁵

Fig. 1 shows the relative weights on stocks in the long/short strategy under the Frazzini and Pedersen (2014) rank-weighting procedure, and in a standard *equal-weighted* portfolio that holds the top fraction θ of all stocks, while shorting the bottom θ (in the figure θ is 1/3). The horizontal axis is the cross-sectional rank of the sorting variable, r , from zero to one. Normalizing so that the total weight in each portfolio is one, the weights on the stock with rank r in the two portfolios shown in the figure

² These data can be found at <https://www.aqr.com/Insights/Datasets/>

³ We follow Baker et al. (2011) in starting our sample at the beginning of 1968, a common start date for the literature on “defensive” strategies. Extending the sample beyond 50 years does not change any qualitative conclusions, but does preclude the use of the Fama and French (2015) profitability and investment factors in the asset pricing tests, as these are only available after June 1963.

⁴ A detailed comparison of the three different versions of BAB is provided in Appendix A.

⁵ There is no reason that a rank-weighting procedure has to ignore market capitalizations. A rank- and capitalization-weighted scheme retains the spirit of rank-weighting, by over-weighting extreme observations, while simultaneously creating portfolios that are largely representative of the size distribution observed in the market. Under this procedure the weight on stock i with signal rank r_i is equal to

$$w_i = \begin{cases} \frac{(r_i - r_{\text{median}})\text{MarketCap}_i}{\sum_{\{j|r_j < r_{\text{median}}\}} (r_j - r_{\text{median}})\text{MarketCap}_j} & \text{if } r_i < r_{\text{median}} \\ \frac{(r_i - r_{\text{median}})\text{MarketCap}_i}{\sum_{\{j|r_j > r_{\text{median}}\}} (r_j - r_{\text{median}})\text{MarketCap}_j} & \text{if } r_i > r_{\text{median}} \end{cases}$$

¹ Li et al. (2019) also look at the impact of non-standard methods, analyzing their impact on the performance of liquidity strategies.

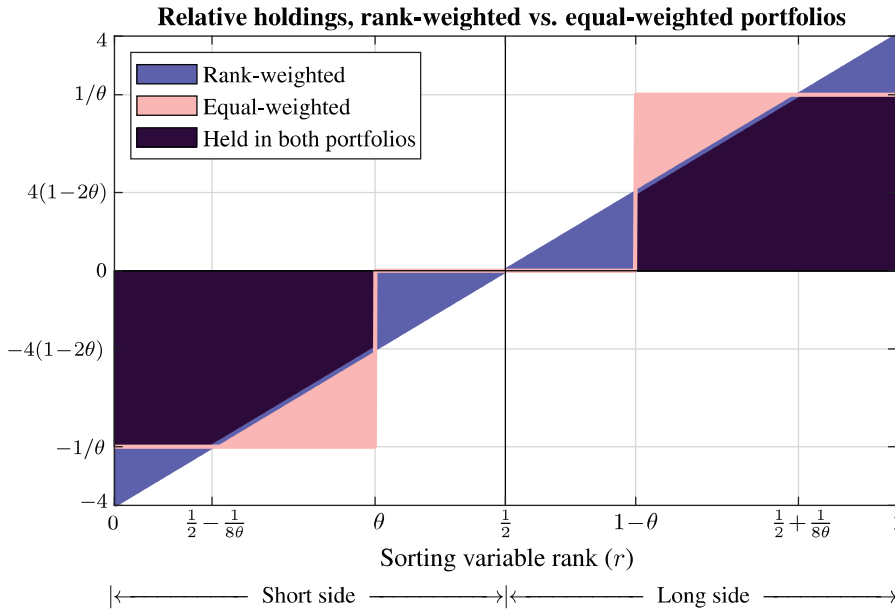


Fig. 1. The figure shows the weight put on stocks (long and short sides) for the [Frazzini and Pedersen \(2014\)](#) rank-weighting procedure and the extreme portfolios from a simple equal-weighted portfolio sort (top and bottom fraction θ of all names; $\theta = 1/3$ shown here). Weights are shown as a function of the signal rank $r \in [0, 1]$, and are normalized so that the (absolute) holdings on the long and short sides total one.

are:

$$W_{\text{rank-weighted}}(r) = 8\left(r - \frac{1}{2}\right)$$

$$W_{\text{equal-weighted}}(r) = \begin{cases} \frac{1}{\theta} & \text{if } r > 1 - \theta \\ -\frac{1}{\theta} & \text{if } r < \theta \\ 0 & \text{otherwise.} \end{cases}$$

A useful heuristic that measures how hard a portfolio “tilts” toward a given strategy is the portfolio’s weighted average sorting variable rank. For the long side of the rank-weighted strategy the weighted average sorting variable rank is

$$\int_{\frac{1}{2}}^1 W_{\text{rank-weighted}}(r) r dr = \int_{\frac{1}{2}}^1 8\left(r - \frac{1}{2}\right) r dr = 5/6.$$

By symmetry the weighted average sorting variable rank on the short side is $1/6$. For the long side of the equal-weighted portfolio sort, the average sorting variable rank is midway between the lower threshold for inclusion in the “high” portfolio ($1 - \theta$) and the maximal possible value (one), so is $1 - \theta/2$, and by symmetry the weighted average sorting variable rank on the short side is $\theta/2$. Given this, the average ranks for the portfolios from a standard equal-weighted quantile sort equals those from the rank-weighting procedure when $\theta = 1/3$, which informed our particular choice for θ in [Fig. 1](#). That is, we should expect that the high-minus-low rank-weighted strategy “tilts” toward whatever characteristic it is designed to capture about as hard as the equal-weighted strategy that buys the top third of all stocks and shorts the bottom third of all stocks on the basis of the same sorting variable.

The actual holdings of the rank-weighted and equal-weighted strategies also exhibit a high degree of commonality. The difference in the holdings of the “high” portfolios from the two different procedures is the area of the trian-

gle above the sloping line in the right half of [Fig. 1](#) (specifically, this is the holdings of the equal-weighted portfolio not held by the rank-weighted portfolio; the two smaller triangles in the right half of the figure together are the holdings of the rank-weighted portfolio not held by the equal-weighted portfolio). The difference in holdings is thus given by

$$\frac{1}{2} \times \underbrace{\left(\left(\frac{1}{2} + \frac{1}{8\theta}\right) - (1 - \theta)\right)}_{\text{Base}} \times \underbrace{\left(\frac{1}{\theta} - 4(1 - 2\theta)\right)}_{\text{Height}} = \left(\frac{1}{4\theta} - 1 + 2\theta\right)^2.$$

When $\theta = 1/3$ this is $(5/12)^2 = 17.4\%$. That is, the equal-weighted portfolio that holds the top third of all stocks and the rank-weighted portfolio hold almost 83 cents of every dollar in common.⁶

Given the high degree of commonality in holdings between the rank-weighted strategy and the equal-weighted strategy that buys and shorts the upper and lower thirds of all stocks, and the fact that the two different constructions yield portfolios with the same weighted-average rank for the sorting variable, it would be quite surprising if the two procedures resulted in significantly different strategy performance. In practice the results are not surprising.

[Fig. 2](#) shows the performance of BAB (solid line), which is based on rank-weighted portfolios, and equal-weighted

⁶ Common holdings are maximized when $\frac{d}{d\theta} \left(\frac{1}{4\theta} - 1 + 2\theta\right)^2 = 0$, so when $\theta = \sqrt{2}/4 = 35.4\%$. For this threshold the difference in holdings for the rank-weighted and equal-weighted portfolios is $3 - 2\sqrt{2} = 17.2\%$. This is only slightly lower than for the portfolio that holds the upper third of all stocks, which is unsurprising—moving the threshold from the upper 35.4% to the upper 33.3% is a small move off the first order condition for portfolio holdings’ commonality.

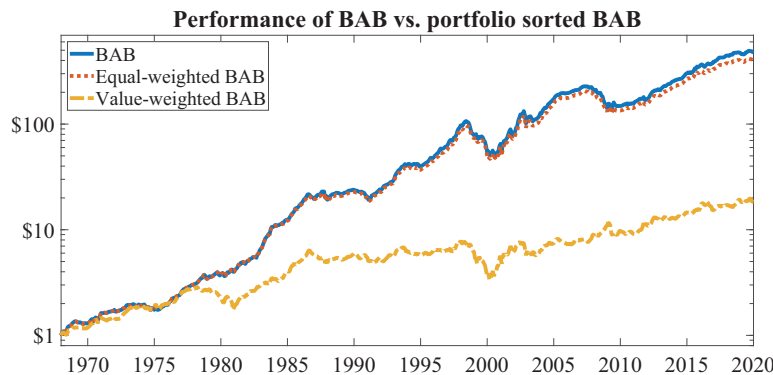


Fig. 2. BAB vs. portfolio sorted BAB. The figure shows the performance of our replication of BAB (solid line), and two variations on BAB, built using the same methods used to construct BAB, but based on quantile-sorted underlying low and high beta portfolios. Equal-weighted BAB (dotted line) is based on equal-weighted portfolios that hold the top and bottom thirds of stocks selected on the basis of the same beta estimates used when constructing the rank-weighted portfolios underlying BAB, while value-weighted BAB (dot-dashed line) is based on the value-weighted versions of these same portfolios. The sample covers January 1968 through December 2019.

BAB (dotted line), which is constructed identically to BAB except that it is based on equal-weighted portfolios that hold the top and bottom thirds of stocks selected on the basis of the same beta estimates used when constructing the rank-weighted portfolios underlying BAB. The figure also shows a value-weighted BAB (dot-dashed line), which is constructed identically except that it is based on value-weighted versions of the same beta portfolios underlying equal-weighted BAB.

The figure shows that BAB and equal-weighted BAB have almost identical performance. In fact, the two strategies are 99.6% correlated at the monthly frequency, and earn statistically indistinguishable average returns. That is, the rank-weighting procedure of [Frazzini and Pedersen \(2014\)](#), while more complicated and more difficult to implement, is essentially identical to a simple equal-weighted portfolio sort.⁷ This close correspondence is not limited to strategies based on beta portfolios. [Appendix B](#) shows that rank-weighted value and momentum are also basically identical to their equal-weighted portfolio-sorted counterparts.

While the rank-weighting procedure yields portfolio performance that is essentially indistinguishable from simple equal-weighted quantile sorting, it is nevertheless the single largest driver of BAB's astonishing performance. This can be seen by contrasting the performance of BAB (or equal-weighted BAB) with that of value-weighted BAB. Value-weighted performance is generally more interesting, because it more accurately reflects what investors can achieve in practice. The performance of the value-weighted version of BAB is much less impressive. Value-weighted

BAB earns significant positive excess returns over the sample, 55 bps/month with a t-statistic of 3.47, but its Sharpe ratio is less than half as large as BAB's (0.48 compared to 1.07).⁸ It also earns most of these returns by tilting strongly to profitability and investment. The strategy has loadings of 0.44 and 0.55 on the Fama and French profitability and investment factors RMW and CMA, respectively, with t-statistics of 6.44 and 5.32. The strategy's alpha relative to the Fama and French five-factor model is consequently only 24 bps/month, and insignificant (t-statistic of 1.68).⁹

2.2. Hedging by leveraging versus simple direct hedging

In their second major deviation from standard factor construction techniques, [Frazzini and Pedersen \(2014\)](#) attempt to make BAB market-neutral using leverage. They scale the underlying portfolios by their predicted betas, with the intention that the leveraged, low beta portfolio and the deleveraged, high beta portfolio each individually have betas of one, and thus net to zero in the long/short strategy.¹⁰

⁸ This is similar to the 51 bps/month return spread [Frazzini and Pedersen \(2014\)](#) report for "value-weighted BAB" in Table B9 of their online appendix. The high and low beta portfolios underlying [Frazzini and Pedersen \(2014\)](#) value-weighted BAB are each constructed as equal-weighted averages of small and large capitalization strategies, along the lines of how [Fama and French \(1993\)](#) build HML. Specifically, the high beta portfolio is a 50/50 combination of the value-weighted portfolios that hold only high beta stocks (top 30%) with either above or below median NYSE market capitalizations, and the low beta portfolio is constructed similarly. [Frazzini and Pedersen \(2014\)](#) "value-weighted BAB" by construction consequently puts half its weight on small stocks that make up on average less than 11% of total market capitalization, and thus cannot accurately be described as "value-weighted."

⁹ Its alpha relative to the six-factor model that also includes momentum is even lower, 14 bps/month with a t-statistic of 0.96. Detailed analysis of the performance of this strategy, and all the strategies considered here, is provided in [Appendix C](#).

¹⁰ This procedure is biased, because of Jensen's inequality and the fact that betas are estimated with noise. If zero is in the continuous support of a portfolio's potential beta estimate, then expected leverage (inverse of the estimated beta) is unbounded, and so is the leveraged portfolio's

⁷ [Han \(2019\)](#) compares the performance of the dollar-neutral strategy that buys and sells the rank-weighted low and high beta portfolios, respectively, to that of the long/short strategy that buys and sells equal-weighted portfolios that hold stocks with below and above median beta. The spread in the weighted average beta rank between this equal-weighted strategy's high and low portfolios is 1/2, only three-quarters of the 2/3 spread for the rank-weighted strategy, or the equal-weighted strategy based on the top and bottom thirds of stocks. The equal-weighted strategy studied by [Han \(2019\)](#) consequently hedges roughly 75% of the rank-weighted low-minus-high beta strategy's returns.

The simple, standard alternative is to buy the market in proportion to the underlying low-minus-high beta strategy's short market tilt, financing the long market position through borrowing. While conceptually it makes sense to hedge using the value-weighted market portfolio, the strategy that results from doing so, "directly hedged BAB," is quite dissimilar to BAB. This is because BAB is constructed by hedging an underlying dollar-neutral low-minus-high beta strategy by buying rank-weighted portfolios, a fact apparent in the following identity for BAB's returns:

$$\begin{aligned}
 r_{\text{BAB}} &= \underbrace{\frac{r_L - r_{rf}}{\beta_L}}_{\text{Leveraged, rank-weighted low beta portfolio}} - \underbrace{\frac{r_H - r_{rf}}{\beta_H}}_{\text{Unleveraged, rank-weighted high beta portfolio}} \\
 &= \underbrace{(r_L - r_H)}_{\text{Dollar-neutral low-minus-high beta strategy underlying BAB}} + \underbrace{((\beta_L^{-1} - 1)r_L + (1 - \beta_H^{-1})r_H)}_{\text{Long position in rank-weighted equity used to hedge the underlying low-minus-high beta strategy}} \\
 &\quad - \underbrace{(\beta_L^{-1} - \beta_H^{-1})r_{rf}}_{\text{Borrowing used to finance the hedge}}.
 \end{aligned}$$

where r_L , r_H , and r_{rf} are the returns to the low beta portfolio, high beta portfolio, and "risk-free" short-term treasuries, respectively, and β_L and β_H are the estimated betas of the low and high beta portfolios.¹¹ The rank-weighted portfolios are, as we saw previously, almost indistinguishable from equal-weighted portfolios, and the combination of rank-weighted low and high beta portfolios used to hedge the underlying low-minus-high beta strategy is

true expected beta. More generally, suppose that a portfolio's true beta is measured with noise, $\hat{\beta} = \beta(1 + \epsilon)$ where ϵ is mean-zero proportional estimation error. Then the expected beta of the portfolio held in inverse proportion to its estimated beta is

$$E\left[\frac{\beta}{\hat{\beta}}\right] = E\left[\frac{1}{1 + \epsilon}\right] = 1 + \sum_{i=2}^{\infty} E[(-\epsilon)^i],$$

where the summation starts at $i = 2$ because $E[\epsilon] = 0$. That is, the expected beta of a portfolio scaled by the inverse of its estimated beta differs from one by the difference between the sums of the even and the higher odd central moments of the noise with which beta is estimated. Under the reasonable assumption that the estimation error is symmetric (or nearly symmetric), the odd moments are zero (or close to zero), and the expected bias is simply (roughly) the sum of the even central moments. In this case, scaling a portfolio by the portfolio's estimated beta yields a strategy that has an expected beta greater than one.

¹¹ Liu et al. (2018) use this same decomposition of BAB's returns. They show that the long hedging portfolio contributes positively to BAB's alpha among stocks with positive expected alpha, but negatively among stocks with negative expected alpha, where the positive and negative alpha stocks are selected using 11 variables associated with prominent return anomalies. It is difficult to compare their results to those of Frazzini and Pedersen (2014) directly, however, because the Liu et al. (2018) version of BAB is not a true replication of the Frazzini and Pedersen (2014) BAB. Liu et al. (2018) report a three-factor alpha on BAB of 0.36 bps/month, with a t-statistic of 3.47, over their sample from January 1963 to December 2013. The original BAB series studied by Frazzini and Pedersen (2014) has a three-factor alpha twice as high, 72 bps/month with a t-statistic of 5.66, over the sample starting January 1963 through its end in March 2013 (BAB performed strongly over the last nine months of 2013, biasing this result down relative to those reported in Liu et al. (2018)). Both the BAB series currently maintained by AQR and our replication of BAB have even higher information ratios relative to the three-factor model.

more similar to the equal-weighted market than it is to the value-weighted market.

We consequently construct two directly hedged versions of BAB, one hedged by buying the value-weighted market in proportion to the observed short market tilt of the underlying low-minus-high beta strategy, and one hedged by buying the equal-weighted market portfolio in proportion to the underlying low-minus-high beta strategy's observed beta on this portfolio.¹² The version constructed using equal-weighted hedging is far more similar to BAB than the version suggested by BAB's underlying theory, which is hedged using the value-weighted market (monthly correlation of 90.2% versus 71.6%).

Fig. 3 shows the performance of our replication of BAB (dotted line), and the versions of BAB hedged by directly buying the value-weighted market (dot-dashed line) or the equal-weighted market (solid line). Because the version hedged by buying the equal-weighted market is in some sense better hedged than BAB, realizing a sample volatility of only 9.5% compared to the 11.8% observed on BAB, all three strategies are scaled to run at the same sample volatility as BAB. The figure shows that the version of BAB directly hedged by buying the equal-weighted market, which is highly correlated with BAB, realized an even higher Sharpe ratio than BAB (1.25 versus 1.07). The version hedged by buying the value-weighted market, which is more consistent with the theory underlying BAB, had much weaker performance, realizing a Sharpe ratio of "only" 0.81.

While Frazzini and Pedersen (2014) leveraging procedure does not yield stronger performance than directly beta-hedging the underlying low-minus-high beta strategy with the equal-weighted market, it is nevertheless crucial, like rank-weighting, for delivering BAB's astonishing performance. Hedging with equal-weighted portfolios instead of value-weighted portfolios contributes significantly to performance, and while not explicit in Frazzini and Pedersen (2014), the leveraging procedure is the backdoor through which the paper implements equal-weighted hedging. The leveraging procedure uses the underlying portfolios to do the hedging, and in BAB's case these are rank-weighted, and thus almost indistinguishable from equal-weighted portfolios. Using these portfolios instead of the market to hedge the underlying strategy's short market tilt drives BAB's strong paper performance.

3. Implementing BAB

Frazzini and Pedersen (2014) rank-weighting procedure creates underlying beta portfolios that are almost indistin-

¹² We directly estimate these betas by regressing the underlying low-minus-high beta strategy's post-formation realized returns onto market excess returns. This avoids the errors-in-variables problem inherent in the procedure Frazzini and Pedersen (2014) employ, which estimates portfolio betas as the holdings-weighted average beta estimates of the stocks they hold, using the same stock beta estimates employed in portfolio formation. Our regressions use one year of daily returns, and attempt to account for the nonsynchronous trading problem of Scholes and Williams (1977) using the correction of Dimson (1979), by calculating beta as the sum of the slope coefficients on the market's contemporaneous return and its one- and two-day lagged returns.

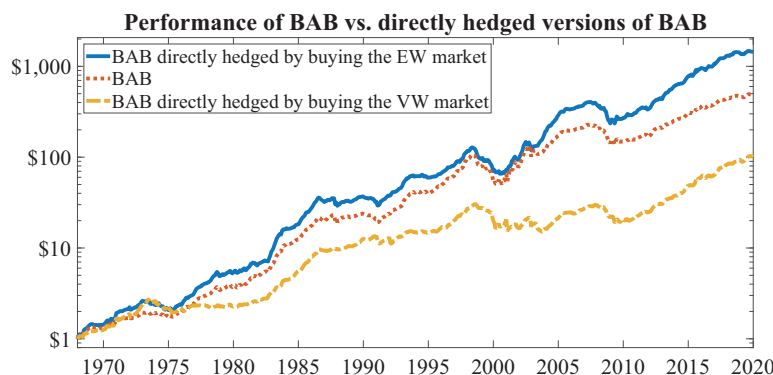


Fig. 3. BAB vs. directly hedged versions of BAB. The figure shows the performance of our replication of BAB (dotted line), and two strategies based on the same low-minus-high beta strategy underlying BAB, but hedged directly with debt-financed positions in the market portfolio. One of these is hedged by buying the equal-weighted market (solid line), while the other is hedged by buying the value-weighted market (dot-dashed line). Because the version directly hedged by buying the equal-weighted market naturally has a significantly lower volatility (9.5%), the directly hedged versions are leveraged to run at the same sample volatility as BAB (11.8%). The sample covers January 1968 through December 2019.

Table 1

BAB Holdings, By NYSE Size Decile.

The table reports, for several strategies, the time-series average holdings of stocks from each market capitalization decile formed using NYSE breaks, as a percent of each dollar invested. The strategies are the market portfolio, the beta portfolios from the low-minus-high beta strategy underlying BAB, and the long and short sides of BAB itself (i.e., the leveraged low beta portfolio and the deleveraged high beta portfolios). The table also reports the market capitalization of the smallest firm in each NYSE size decile at the end of the sample. The sample covers January 1968 through December 2019.

Decile	Smallest firm, end 2019 (\$)	Portfolio market capitalization by NYSE size decile (%)				
		Market	Rank-weighted beta portfolios		BAB	
			Low- β	High- β	Long-side	Short-side
Mega	32.8 B	59.4	3.1	4.4	4.2	3.4
2	14.1 B	13.6	3.2	4.6	4.5	3.5
3	7.54 B	7.8	3.1	5.1	4.5	3.7
4	4.76 B	5.1	3.2	5.4	4.7	4.0
5	2.94 B	3.7	3.5	6.0	5.2	4.3
6	1.91 B	2.9	4.1	6.9	6.1	5.0
7	1.16 B	2.3	4.7	8.0	7.3	5.8
8	621 M	1.8	6.0	9.6	9.5	7.0
Micro	268 M	1.6	9.8	13.4	15.8	9.7
Nano	0.94 M	1.7	59.2	36.6	94.6	26.1
Total		100.0	100.0	100.0	156.4	72.5

guishable from equal-weighted portfolios, and [Frazzini and Pedersen \(2014\)](#) leveraging procedure uses these same portfolios to hedge the strategy. As a result, the strategy dramatically over-weights the smallest, most illiquid stocks, making it infeasible in practice.

To understand how acute BAB's equal-weighting problem is, we can look at the actual weights the strategy puts on stocks from different size "universes." [Table 1](#) does this, where the size universes are size deciles formed using NYSE breaks. These divide the market into ten buckets based on market capitalization, where the breaks are set so that, by construction, 10% of NYSE stocks fall into each. NASDAQ and AMEX stocks tend to be smaller, so are more likely to fall into the smaller stock portfolios. To provide a sense of the kinds of stocks held in each universe, the second column of the table shows the smallest stock held in each at the end of the sample. For example, at the end of the sample "large stocks," under the [Fama and](#)

[French \(2008\)](#) definition of those with above NYSE median market capitalization (deciles 1–5), are those with market caps in excess of about \$3 billion. "Micro-caps" (deciles 9 and 10 using NYSE breaks) at the end of the sample are those with capitalizations under \$621 million, and "small" are those in between.

To provide a benchmark against which to compare BAB's holdings, the third column provides the time-series average market capitalization of each size decile as a fraction of the total market portfolio. On average large caps (top five deciles) account for almost 90% of the market, with the top decile alone accounting for almost 60%. The bottom three deciles on average each account for less than 2% of the total market.

These numbers contrast sharply with where BAB invests. The high beta portfolio underlying the short side puts on average almost 40 cents of every invested dollar into the nano-cap stocks in the smallest size decile,

Table 2

BAB Turnover, By NYSE Size Decile.

The table reports BAB's average annual turnover in each NYSE market capitalization decile. Turnover is calculated as the average of purchases and sales, and expressed as a percentage of the long-only investment (beta portfolios), or as a percentage of the total investment in the long/short strategy (BAB). Annual turnover is 12 times monthly turnover. The sample covers January 1968 through December 2019.

Decile	Average annual turnover in each NYSE size decile as a percentage of the long-only investment (beta portfolios) or as a percentage of the investment in the long/short strategy (BAB)				
	Rank-weighted beta portfolios		BAB		
	Low- β	High- β	Long-side	Short-side	Total
Mega	1.9	3.3	2.7	2.8	5.5
2	2.0	3.7	3.0	3.1	6.0
3	2.1	4.1	3.2	3.4	6.7
4	2.3	4.5	3.5	3.7	7.2
5	2.5	5.1	3.9	4.3	8.1
6	3.0	6.0	4.7	5.0	9.8
7	3.6	7.1	5.8	6.0	11.8
8	4.6	9.1	7.7	7.6	15.2
Micro	7.7	13.3	12.9	11.0	23.9
Nano	50.2	45.3	84.4	35.7	120.1
Total	79.8	101.4	131.8	82.6	214.4

which makes up on average only 1.7% of the market. The low beta portfolio underlying the long side invests even more aggressively in these nano-caps, putting on average almost 60 cents of every invested dollar into these stocks. BAB itself, because it leverages this low beta portfolio on average by more than 50%, puts an average of 94.6 cents of every dollar into low beta stocks in the smallest decile, while shorting 26.1 cents of high beta stocks there. On net, for every dollar invested in BAB, the strategy takes equity positions that on average exceed \$1.20 in these stocks that make up the bottom 1.7% of the market.¹³ This presents significant implementation issues, because the smallest stocks have limited capacity and are expensive to trade.

3.1. The cost of trading BAB

BAB dramatically overweights nano- and micro-cap stocks that are expensive to trade, making it far less profitable in practice than on paper. We can get a sense of how difficult BAB is to trade by calculating the strategy's average turnover in each NYSE size decile. Table 2 gives this turnover, as a fraction of the total capital invested in the strategy. Overall, BAB entails significant trading, and like its holdings, this trading is heavily concentrated in the smallest stocks. Each dollar invested in the strategy is associated on average with more than two dollars of annual turnover (214.4% per year), with \$1.32 of round trip trading on the long side and \$0.83 of round trip trading on the short side. While this turnover is not especially high (roughly three times the turnover of a value strategy, but only a third the

turnover of momentum), more than half of this trading is in nano-cap stocks in the smallest NYSE size decile (120.1% per year), and these stocks are expensive to trade.

This turnover translates into significant transaction costs, averaging 58 bps/month, calculated using the methodology of Novy-Marx and Velikov (2016).¹⁴ These costs dramatically reduce the performance a BAB investor would have realized in practice. Fig. 4 shows the cumulative returns to BAB after accounting for these transaction costs (BAB net, solid line), and for comparison includes BAB's gross performance (dotted line). The figure shows that while accounting for transaction costs reduces BAB's profitability more than 55%, the strategy still earns significant average net returns, 47 bps/month with a t-statistic of 3.35. BAB's generalized net alpha (Novy-Marx and Velikov, 2016) relative to the Fama and French five-factor model, however, is only 17 bps/month and insignificant (t-statistic of 1.32; see Appendix C for details). This generalized alpha measures the extent to which a test asset could have improved the ex post mean-variance efficient portfolio, accounting for the costs of trading both the test asset and the explanatory factors. BAB tilts strongly toward profitability and investment, tilts for which it is fairly compensated. The strategy is a backdoor way to getting exposure to these factors, which explain its significant average returns. The modest five-factor generalized alpha reflects the fact that an investor already trading profitability and investment could not in practice have realized significant performance improvements by additionally trading BAB.¹⁵

¹³ If we define nano-cap more narrowly, as the bottom 1% of market capitalization, BAB's equity position in nano-caps averages "only" \$1.05. More than a third of this is in stocks that make up the bottom 0.1% of the market, firms with end of sample market capitalizations under \$106 million.

¹⁴ These costs averaged only 17 bps/month over the last ten years of the sample. Despite this BAB was less profitable to trade over this period, because it also realized significantly lower gross returns.

¹⁵ Replicating BAB excluding the bottom NYSE decile, which makes up on average only 1.7% of total market capitalization, significantly reduces the cost of trading BAB, from 58 bps/month down to 22 bps/month (and only 9 bps/month over the last ten years of the sample). This has almost

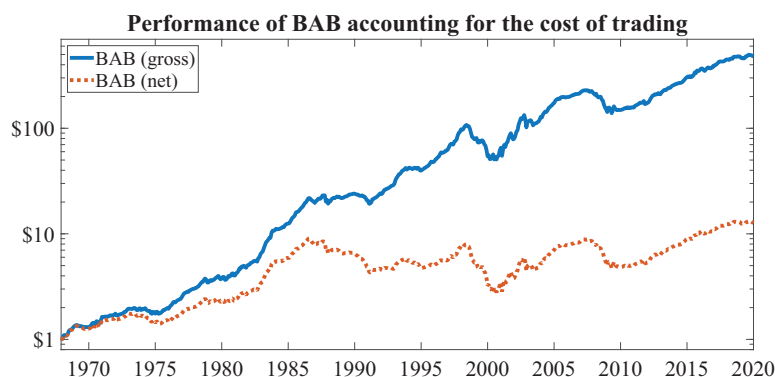


Fig. 4. BAB performance accounting for the cost of trading.

The figure shows the performance of BAB net of transaction costs (solid line). For comparison the figure also includes the strategy's performance ignoring transaction costs (dotted line). Transaction costs are estimated following Novy-Marx and Velikov (2016). The sample covers January 1968 through December 2019.

4. Impact of Frazzini and Pedersen (2014) non-standard beta estimation procedure

The third major non-standard procedure Frazzini and Pedersen (2014) use when constructing BAB is a novel method for estimating individual stocks' betas, which are a crucial input when constructing BAB. Cross-sectionally, beta estimates are used to construct the high and low beta portfolios underlying BAB. In the time-series, beta-estimates are used to hedge the market loadings of these portfolios. We show here that while the FP-beta estimates perform roughly as well as other common beta estimators cross-sectionally, generating similar post-formation spreads in beta-sorted portfolios, they are strongly biased in ways that vary predictably over time. This time-series bias has little impact on BAB's average returns (see Appendix D), but drives empirical results that Frazzini and Pedersen (2014) present in support of their theory.

4.1. Cross-sectional performance of the FP-beta estimator

BAB's construction begins with underlying portfolios designed to have a large difference in market loadings. A reasonable metric for the cross-sectional performance of the beta estimates used to construct these portfolios

is consequently the ability of the beta estimates to successfully generate a large realized beta spread between the high and low beta portfolios. We measure realized beta spreads using full sample CAPM regressions employing monthly returns, which avoids nonsynchronous trading issues that plague estimates calculated using daily data, issues that are explored in detail in Appendix E.

Fig. 5 shows full-sample realized beta spreads between high and low beta portfolios, created using six different beta estimators: OLS estimates calculated using contemporaneous one-day and three-day returns, the estimators of Vasicek (1973) and Welch (2019), a Dimson (1979) estimator calculated by combining the loadings on contemporaneous and one-day lagged daily market returns, and the FP-beta estimator (the first five all use one year of daily data, while the last uses a longer five-year window to calculate stocks' market correlations). The figure shows these realized high-minus low beta spreads when stocks are sorted on the basis of these beta estimators into two, three, five, and ten portfolios, constructed using name breaks and monthly rebalancing.

All six beta estimators perform similarly, with the best performing estimator (Dimson) generating realized beta spreads roughly 10% higher than the worst performing estimator (Welch) across the spectrum of sorts (0.54 versus 0.48 when employing only two portfolios; 1.21 vs 1.06 for the decile sort). The estimators that make no attempt to account for nonsynchronous trading (OLS from one-day returns, Vasicek, and Welch) always perform worse than those that do (OLS from three-day returns, Dimson, and FP). Including the betas on one-day lagged market performance increases the beta spreads more than employing three-day overlapping returns. Estimates that use three-day overlapping returns to control for nonsynchronous trading do increase spreads relative to methods that simply employ contemporaneous one-day returns. Frazzini and Pedersen (2014) procedure of combining correlations estimated using five years of three-day overlapping returns with volatilities estimated using one year of daily returns increases the realized beta spreads less, however, than simply estimating betas directly using OLS and one year of overlapping three-day returns.

no impact on net performance, however, because of commensurate reductions in gross performance. BAB constructed excluding these nano-cap stocks earns net returns of 46 bps/month with a t-statistic of 3.38, and has a generalized alpha relative to the Fama and French five-factor model of 15 bps/month (t-statistic of 1.31), net performance almost identical to that observed on BAB. Similarly, rank- and capitalization-weighting BAB (details provided in Footnote ⁵), so that the strategy holds stocks with a size distribution more representative of the broad market, has little impact on the performance an investor would have actually realized. This "implementable" version of BAB costs less than one seventh as much as BAB to trade, only 8 bps/month (4 bps/month over the last ten years of the sample). Despite this, its average net returns are slightly lower, 42 bps/month with a t-statistic of 2.82, because when the strategy holds stocks with a size distribution similar to the market's it earns gross returns that are only half as large as Frazzini and Pedersen (2014) purely rank-weighted BAB. This version's generalized alpha relative to the Fama and French five-factor model is also insignificant, 19 bps/month with a t-statistic of 1.40.

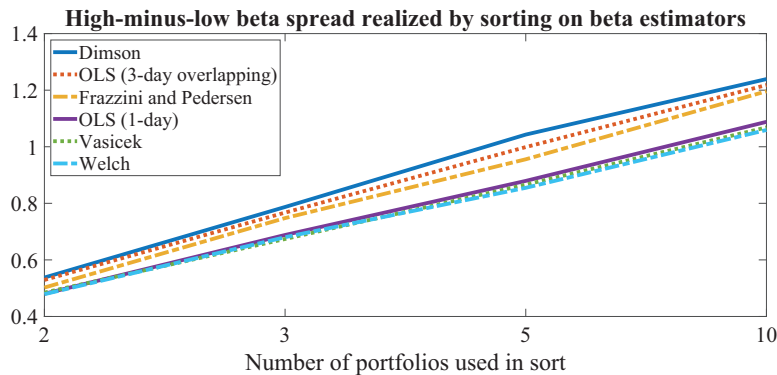


Fig. 5. Beta spread realized by sorting on different beta estimators. The figure shows the beta spread realized by sorting on individual stocks' beta estimates calculated using six different estimating procedures: Ordinary Least Squares (OLS) calculated using one-day and three-day returns, Dimson-corrected estimates that employ one lag, the estimators of Vasicek (1973) and Welch (2019), the estimator used by Frazzini and Pedersen (2014), and an OLS estimate calculated using one year of daily data that employs three-day overlapping returns. The figure shows the realized high-minus-low beta spreads for sorts into two, three, five, and ten portfolios. Spreads are estimated as full-sample market loadings from market-model regressions using monthly returns. The sample covers January 1968 through December 2019.

Overall, the FP-betas perform reasonably well, though less well than the common Dimson estimator, at generating large realized beta spreads between beta sorted portfolios. Appendix E provides additional supporting evidence, exploring the extent to which the different measures contain marginal information relative to each other, evidence that again generally favors the Dimson estimator. It also shows how all of the beta estimators can be adjusted for market capitalization, which is negatively correlated with the extent to which they are biased downward by nonsynchronous trading issues, so that they generate larger realized beta spreads.

4.2. Time-series bias in the FP-beta estimator

While the FP-betas perform reasonably well in the cross-section, they are biased in the time-series, and this bias has important implications for the empirical results that Frazzini and Pedersen (2014) present in support of their underlying theory. It is easy to calculate the value-weighted average bias in the FP-betas, or of any beta estimator, because the true value-weighted average beta of all the stocks that make up the market is always exactly one. The extent to which the value-weighted market average of any beta estimate differs from one consequently provides a simple summary statistic of the estimator's value-weighted average bias.

Fig. 6 shows the value-weighted market average of the FP-beta estimator (solid blue line), together with value-weighted market averages of two other commonly employed estimators, simple OLS calculated using one year of contemporaneous daily returns (dotted yellow line), and the Dimson estimator, which performed strongest in the cross-sectional analysis, shrunk toward one using the same weights employed by Frazzini and Pedersen (2014).¹⁶ All

the estimators show some value-weighted average bias. For example, the value-weighted averages of both the OLS and Dimson estimators become biased upward in 2001, as high beta stocks outperform the market as a whole, increasing their weight in the value-weighted average estimate, and then suffer precipitous declines in the following bear market, as the high beta stocks again lead low beta stocks and shed value much more quickly (see, for example, Daniel and Moskowitz, 2016). The value-weighted average bias in these common measures is generally dwarfed, however, when compared to the value-weighted average bias in the FP-beta estimates. The time-series standard deviation in the value-weighted average bias in the FP-beta estimator is 9.4%, but only 2.9% for the OLS estimator, and 1.9% for the Dimson estimator. The value-weighted FP-beta estimator is also biased up on average in the time-series, with a mean 4.6% above one, while the value-weighted averages of the other two estimators have almost no average bias, with time-series means of 1.007 and 1.005, respectively.

Table 3 provides direct evidence that time-series variation in the value-weighted average bias in the FP-beta estimator predicts the extent to which portfolios hedged using FP-betas are mis-hedged. The table reports results from time-series regressions of the monthly returns to portfolios market-hedged using FP-beta estimates, i.e., regressions of the form

$$r_{\text{portfolio excess}} - \left(\hat{\beta}_{\text{FP}}^{\text{portfolio}} \times \text{MKT} \right) = \alpha + \beta \text{MKT} + \gamma \left(\hat{\beta}_{\text{FP}}^{\text{MKT}} - 1 \right) \times \text{MKT} + \epsilon.$$

so it does not affect the cross-sectional results presented in the previous section, or change the beta portfolios underlying BAB. It does impact BAB's performance, however, because the degree of shrinkage used for the individual stock betas estimates is inherited by portfolio beta estimates, and is thus a parameter that directly impacts the leverage employed for hedging. This problem is avoided when directly estimating portfolio betas using the portfolios' realized returns.

¹⁶ Frazzini and Pedersen (2014) put a 60% weight on their empirically estimated individual stock betas, shrinking them towards one (the true market-weighted average of individual stocks' market betas) using a 40% weight. This shrinkage does not affect the rank-ordering of beta estimates,

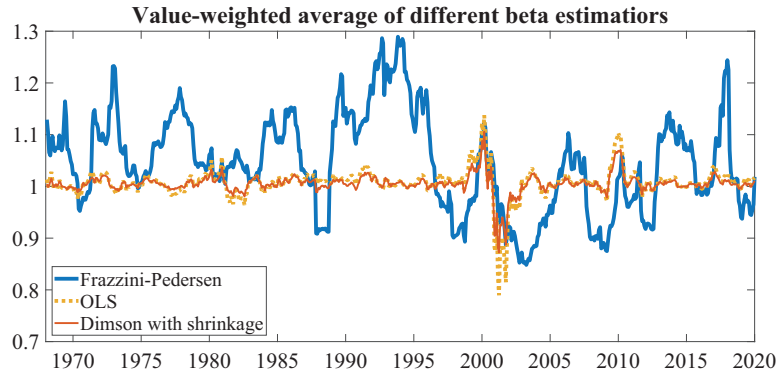


Fig. 6. Value-weighted average of different beta estimators. The figure shows the value-weighted average beta of all the stocks in the market for three beta estimators, simple OLS using one year contemporaneous daily returns, the estimator of Frazzini and Pedersen, and a one year OLS estimate Dimson corrected using one daily lag and the same degree of shrinkage employed by Frazzini and Pedersen (2014) (0.6 on the estimate and 0.4 on one). The true value-weighted average beta of all the stocks in the market is always exactly one, so the extent to which these averages differ from one quantifies the value-weighted average bias in each beta estimators. The sample covers January 1968 through December 2019.

Table 3

Predicting hedging bias using the average bias in the FP-beta estimates.

The table reports results from time-series regressions of the monthly returns to portfolios hedged by shorting the market in proportion to the value-weighted average beta estimates of the portfolios' holdings at the start of the month,

$$r_{\text{portfolio excess}} - \left(\hat{\beta}_{\text{FP}}^{\text{portfolio}} \times \text{MKT} \right) = \alpha + \beta \text{MKT} + \gamma \left(\hat{\beta}_{\text{FP}}^{\text{MKT}} - 1 \right) \times \text{MKT} + \epsilon.$$

Dependent returns are hedged by subtracting the contemporaneous market returns in proportion to the beginning of the month weighted-average of the FP-beta estimates of the portfolio's holdings, $\hat{\beta}_{\text{FP}}^{\text{portfolio}} \times \text{MKT}$. Explanatory variables are the contemporaneous market excess return, MKT, and the market excess returns interacted with the value-weighted average bias in the market FP-beta estimate at the start of the month, $(\hat{\beta}_{\text{FP}}^{\text{MKT}} - 1) \times \text{MKT}$. Test assets include the market portfolio (both value-weighted and equal-weighted), the portfolios underlying BAB (portfolios rank-weighted on FP-beta estimates), and the 25 Fama and French portfolios sorted on size and book-to-market. The sample covers January 1968 through December 2019.

Panel A: Full results for market portfolios and the portfolios underlying BAB				
	α	β	γ	adj.-R ² (%)
Value-weighted market	0	0	−1	100
Equal-weighted market	0.09	0.18	−1.59	8.2
Low beta	[0.74]	[6.53]	[−5.19]	1.7
	0.56	0.07	−0.57	
High beta	[5.68]	[3.15]	[−2.34]	9.7
	−0.26	0.24	−3.13	
	[−1.48]	[5.84]	[−7.03]	

Panel B: β s and γ s for the 25 Fama and French portfolios sorted on size and book-to-market										
	Slope coefficients					t-statistics				
	Small	2	3	4	Big	Small	2	3	4	Big
Panel B1: β s										
Growth	0.38	0.32	0.23	0.15	−0.05	8.63	9.57	8.60	7.10	−3.24
2	0.24	0.17	0.13	0.10	−0.05	6.17	6.35	6.27	6.15	−4.08
3	0.16	0.10	0.07	0.06	−0.08	4.86	4.31	3.50	3.08	−4.99
4	0.10	0.08	0.04	0.01	−0.07	3.23	3.15	1.69	0.55	−3.62
Value	0.11	0.09	0.04	0.02	−0.12	3.41	3.23	1.46	0.80	−4.60
Panel B2: γ s										
Growth	−1.48	−1.12	−0.80	−0.58	−1.20	−3.01	−3.04	−2.70	−2.55	−7.33
2	−1.12	−1.16	−0.99	−1.06	−0.90	−2.66	−3.87	−4.43	−5.99	−6.16
3	−1.05	−0.88	−1.04	−1.15	−1.11	−3.00	−3.29	−4.66	−5.69	−6.02
4	−0.98	−1.10	−1.14	−1.20	−1.33	−2.94	−4.02	−4.82	−5.59	−5.92
Value	−1.46	−1.36	−1.18	−1.35	−1.07	−4.02	−4.27	−4.07	−4.96	−3.82

Portfolios' returns are hedged by subtracting the contemporaneous market returns in proportion to the beginning of the month weighted-average of the FP-beta estimates of the portfolio's holdings, $\hat{\beta}_{\text{FP}}^{\text{portfolio}} \times \text{MKT}$. Explanatory variables are the contemporaneous market excess return, MKT, and the market excess returns interacted with the

value-weighted average bias in the market FP-beta estimate at the start of the month, $(\hat{\beta}_{\text{FP}}^{\text{MKT}} - 1) \times \text{MKT}$. Test assets include the market portfolio (both value-weighted and equal-weighted), the portfolios underlying BAB (portfolios rank-weighted on FP-beta estimates), and the 25 Fama and French portfolios sorted on size and book-to-market.

The sample covers January 1968 through December 2019.

Panel A shows full regression results for the market portfolios and the high and low beta portfolios underlying BAB. The explanatory variables explain 100% of the variation in the hedged returns of the value-weighted market, by construction, and these results are provided only for reference. The value-weighted average bias in the FP-betas explains all of the mishedging of the market hedged with these same betas, so the slope coefficient on the interaction term is exactly negative one, i.e., the market is over hedged by exactly the bias in the value-weighted average FP-beta estimates. For the equal-weighted market, the direct market loading is positive, 0.18, reflecting average underhedging. Fig. 6 shows that on average there is a positive value-weighted average bias in the FP-beta estimates, but Appendix E provides direct evidence that all the common procedures for estimating beta in daily data have strong downward biases among small cap stocks, resulting from nonsynchronous trading issues. The equal-weighted market has roughly 75% of its holdings in stocks with below NYSE median market capitalization, so it inherits this average downward bias. The slope coefficient on the interaction term is -1.59 , suggesting that on average the time varying mishedging of the equal-weighted market that results from hedging using the FP-betas is almost 60% higher than the mishedging of the value-weighted market. Results for the portfolios underlying BAB also show average underhedging, because the rank-weighting procedure dramatically overweights small cap stocks, and a mishedging from the value-weighted average bias in the FP-beta estimates that is mitigated in the low beta portfolio (slope coefficient on the interaction term of only -0.57), but strongly amplified in the high beta portfolio (slope coefficient on the interaction term of -3.13).

Panel B reports results for some of the most widely used test assets in finance, the 25 Fama and French portfolios sorted on size and book-to-market.¹⁷ To conserve space it shows only slope coefficients and their t-statistics on the direct market loadings, and on the market interacted with the value-weighted average bias in the FP-betas (β and γ in the regression equation respectively). The direct market loadings on the five large cap portfolios are negative, because the FP-betas used to calculate the hedge have a positive time-series mean value-weighted average bias, resulting in some overhedging on average. The direct market loadings on the other 20 portfolios are negative, because of the downward bias in the small cap beta estimates. The slope coefficients on the interaction term are negative for all 25 portfolios, ranging from -0.58 to -1.48 , and always highly significant (though none is significantly different from -1), suggesting that the portfolios are overhedged when the value-weighted average FP-beta estimate is biased upward, and underhedged when the value-weighted average FP-beta estimate is biased downward.

¹⁷ These are created from intersecting independent quintile sorts on market equity and book-to-market, both constructed using NYSE breaks. Portfolio returns are value-weighted, and the portfolios are rebalanced annually at the end of each June.

4.3. Market volatility drives the value-weighted average bias in the FP-betas

Frazzini and Pedersen (2014) estimate stocks' betas (pre-shrinkage) by combining correlations and volatilities, where the authors "use a one-year rolling standard deviation for volatilities and a five-year horizon for the correlation to account for the fact that correlations appear to move more slowly than volatilities" (p. 8). While this justification for their split-frequency estimation procedure may sound reasonable, it is directly responsible for the large value-weighted average bias in their estimator. This can be easily seen in the following two identities for the i th stock's FP-beta,

$$\beta_{FP}^i \equiv \frac{\rho_5^i \sigma_1^i}{\sigma_1^{mkt}} = \begin{cases} \left(\frac{\sigma_5^{mkt} \sigma_1^i}{\sigma_5^i \sigma_1^{mkt}} \right) \left(\frac{\rho_5^i \sigma_5^i}{\sigma_5^{mkt}} \right) = \left(\frac{\sigma_1^i / \sigma_5^i}{\sigma_1^{mkt} / \sigma_5^{mkt}} \right) \beta_5^i \\ \left(\frac{\rho_5^i}{\rho_1^i} \right) \left(\frac{\rho_1^i \sigma_1^i}{\sigma_1^{mkt}} \right) = \left(\frac{\rho_5^i}{\rho_1^i} \right) \beta_1^i. \end{cases} \quad (2)$$

That is, using the upper identity, a stock's FP-beta is its realized trailing five-year beta, estimated directly as the slope coefficient on the market factor from a CAPM regression, times the ratio of the stock's volatility estimated using the last one year and five years of data, scaled by the ratio of the market's volatility estimated at the same one and five year horizons.¹⁸ Alternatively, using the lower identity, a stock's FP-beta is its realized trailing one-year beta, estimated directly as the slope coefficient on the market factor from a CAPM regression, times the ratio of the stock's correlation with the market estimated using the last five years of data to its correlation with the market estimated using just the last one year of data.

These relations have strong time-series implications for the FP-beta estimator. In particular, they imply that the value-weighted average bias in the FP-betas seen in Fig. 6 can be predicted using market volatility. While average individual stock volatility tends to be high in absolute terms when the market is more volatile, it tends to be low at these times when measured as a multiple of market volatility. That is, when σ_1^{mkt} is high, then on average $\sigma_1^i / \sigma_1^{mkt}$ is low. The upper identity in Eq. (2) therefore suggests that when the market is volatile then most stocks' FP-betas will be biased downward relative to those obtained directly from CAPM regressions using the preceding five years of data.¹⁹

The basic volatility facts underlying the predictability in the FP-betas' bias are apparent in Fig. 7. Panel A shows

¹⁸ This is a mechanical relation, so unsurprisingly holds strongly in the data. In a Fama and MacBeth (1973) regression, $\beta_5^i \sigma_1^i / \sigma_5^i$ explains on average 99.0% of the contemporaneous cross-sectional variation in β_{FP}^i , and $\sigma_5^{mkt} / \sigma_1^{mkt}$ explains 99.4% of the time-series variation in the slope coefficient estimates from the cross-sectional regressions.

¹⁹ Alternatively, increases in the average correlation between stocks is the most important driver of increases in market volatility. Unsettled markets consequently tend to occur at times when the typical stock is more correlated with the market, and thus when on average $\rho_1^i > \rho_5^i$. The lower inequality of Eq (2) therefore suggests that on average when market volatility is high then most stocks' FP-betas will also be biased downward relative to their market loadings estimated directly from CAPM regressions using the preceding one year of data.

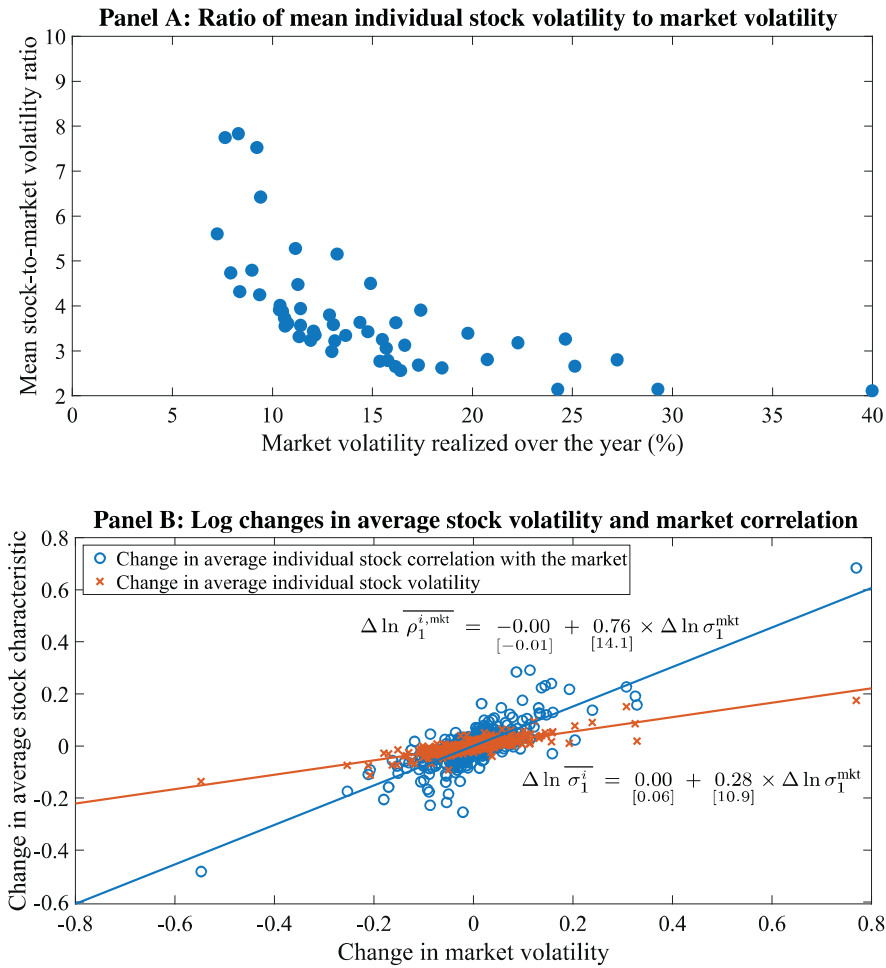


Fig. 7. Individual stock volatilities' and market correlations' relations to market volatility. Panel A shows the average of individual stocks' volatilities realized over each year from 1968 through 2019, as a multiple of market realized volatility in that same year, plotted against the market's realized volatility. Panel B shows the monthly innovations to the average of individual stock volatilities realized over the preceding year ($\Delta \ln \bar{\sigma}_1^i$) and the monthly innovations to the average correlation realized between individual stocks and the market over the preceding year ($\Delta \ln \bar{\rho}_1^{i, \text{mkt}}$), plotted against the monthly innovations to the market volatility realized over the preceding year ($\Delta \ln \sigma_1^{\text{mkt}}$). The figure includes OLS best fit lines for how the average correlation between individual stocks and the market varies contemporaneously with innovations to market volatility, and how average stock volatility varies contemporaneously with innovations to market volatility. The reported t-statistics are calculated using Newey-West Standard errors employing 12 monthly lags.

the average individual stock volatility realized each calendar year from 1968 through 2019, measured as a multiple of, and plotted against, the market volatility realized over the same year.²⁰ In the ten calendar years with the highest realized market volatilities, the realized average individual stock volatility is only 2.71 times as high as the market's volatility; in the ten years with the lowest realized market volatilities that ratio is 5.71. Regressions using the full 52-year sample yield an elasticity of -0.64 . That is, a 1% increase in market volatility is associated on average with a decrease in average individual stock volatility, when measured as a multiple of market volatility, of 0.64%.

The reason for the pattern observed in Panel A is apparent in Panel B, which shows monthly log-changes in the average of the individual stock volatilities realized over

the preceding year ($\Delta \ln \bar{\sigma}_1^i$) and monthly log changes in the average correlation realized between individual stocks and the market over the preceding year ($\Delta \ln \bar{\rho}_1^{i, \text{mkt}}$), both plotted against monthly log changes in the market volatility realized over the preceding year ($\Delta \ln \sigma_1^{\text{mkt}}$). The figure shows that increases in market volatility are associated with increases in average individual stock volatility, but more strongly with increases in the average correlation between individual stocks and the market. In fact, the regressions suggest that on average almost three-quarters of changes in market volatility are driven by changes in correlations, while little more than a quarter are driven by changes in average individual stock volatility.

High market volatility is consequently associated with high average stock correlations, and low average individual stock volatility relative to market volatility. Eq. (2) consequently implies that when market volatility is high then FP-betas tend to be lower, on average, than directly esti-

²⁰ Volatilities are estimated as annualized standard deviations of daily log returns realized over the year.

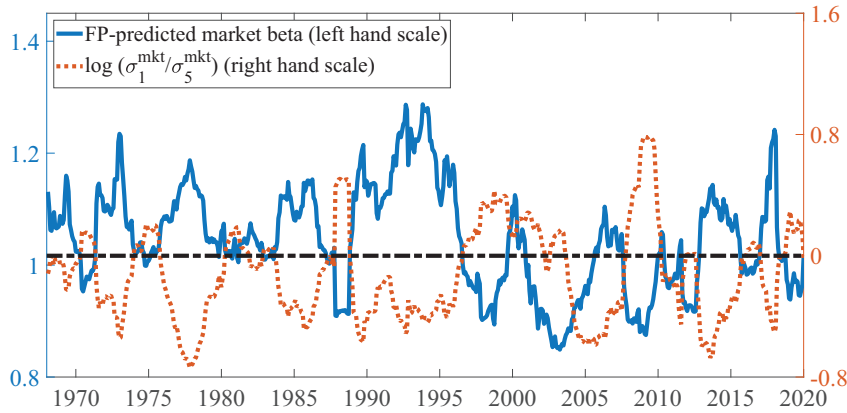


Fig. 8. The value-weighted average bias in the FP-betas' relation to market volatility. The figure shows the predicted market FP-beta (solid line; left hand scale), calculated as the value-weighted FP-beta of the stocks that make up the market portfolio (the procedure Frazzini and Pedersen (2014) use to calculate the betas used to leverage/unleverage the portfolios underlying BAB). It also shows the log market volatility ratio, $\log(\sigma_1^{\text{mkt}}/\sigma_5^{\text{mkt}})$ (dotted line; right hand scale), which is a highly significant time-series predictor of FP-betas. The sample covers January 1968 through December 2019.

mated five-year market betas. Of course the converse is also true, with low market volatility associated with FP-betas that on average exceed market betas.

Fig. 8 confirms this prediction directly. The solid line again shows the value-weighted average FP-beta for the whole market, which quantifies the value-weighted bias in the FP-beta estimator.²¹ The figure shows the strong negative correlation between the value-weighted average bias in the FP-betas and market volatility. The dotted line is the log-market volatility ratio, $\log(\sigma_1^{\text{mkt}}/\sigma_5^{\text{mkt}})$. The value-weighted average FP-beta tends to be strongly biased upward precisely when market volatility is low relative to its longer term average, and biased downward when market volatility is high. Regressing the time-series of the value-weighted average bias in the FP-betas onto the log market volatility ratio yields a slope coefficient of -0.33 with a t-statistic of -4.77 (Newey-West standard errors calculated using 60 monthly lags). The log-market volatility ratio explains 48% of the variation in the predicted market FP-beta, and even more of the variation (60%) in the cross-sectional standard deviation of individual stocks' FP-betas.

²¹ While a portfolio's beta is the value-weighted average beta of its holdings, and Frazzini and Pedersen (2014) estimate the betas used to leverage and deleverage the low and high beta portfolios underlying BAB this way, Eq. (2) implies that this is *not* a valid procedure for FP-betas. To see this, suppose a portfolio holds n stocks with weights given by w^i for $i \in \{1, 2, \dots, n\}$ where $\sum_{i=1}^n w^i = 1$. Then the portfolio's FP-beta is

$$\begin{aligned}\beta_{\text{FP}}^{\text{prt}} &= \left(\frac{\sigma_1^{\text{prt}}/\sigma_5^{\text{prt}}}{\sigma_1^{\text{mkt}}/\sigma_5^{\text{mkt}}} \right) \beta_5^{\text{prt}} = \left(\frac{\sigma_1^{\text{prt}}/\sigma_5^{\text{prt}}}{\sigma_1^{\text{mkt}}/\sigma_5^{\text{mkt}}} \right) \sum_{i=1}^n w^i \beta_5^i \\ &= \sum_{i=1}^n \left(\frac{\sigma_1^{\text{prt}}/\sigma_5^{\text{prt}}}{\sigma_1^i/\sigma_5^i} \right) w^i \beta_{\text{FP}}^i.\end{aligned}$$

This is not equal to $\sum_{i=1}^n w^i \beta_{\text{FP}}^i$, the weighted-average FP-beta estimate. Note this also implies that the directly estimated FP-beta of a long/short strategy is not the difference in the FP-beta estimates of the strategy's long and short sides.

4.4. Implications for the empirical evidence Frazzini and Pedersen (2014) provide in support of their model

The fact that the FP-beta estimates have a large value-weighted average bias that strongly covaries with market volatility has profound implications for the interpretation of the empirical evidence Frazzini and Pedersen (2014) provide in support of their paper's underlying theory. Frazzini and Pedersen (2014) argue that their model “predicts that the betas of securities in the cross section are compressed toward one when funding liquidity risk is high” and claim to show that “consistent with [Frazzini and Pedersen (2014)] Proposition 4, the cross-sectional dispersion in betas is lower when credit constraints are more volatile” (p. 17). In particular, Frazzini and Pedersen (2014) argue that TED-spread volatility is a proxy for funding constraints, and present evidence that beta dispersion, measured by the observed cross-sectional standard deviation, mean absolute deviation, or interquartile range, is significantly lower when TED-spread volatility is high. This evidence only applies to dispersion in FP-betas, however, which as discussed above are strongly biased in predictable ways.

Table 4 evaluates Frazzini and Pedersen (2014) beta compression claims, contrasting the power that TED-spread volatility has predicting dispersion in the biased FP-betas to its lack of power predicting the spread in the high and low beta portfolios' realized market betas. The first specification of Panel A mimics Frazzini and Pedersen (2014) Table 10, investigating the extent to which TED-spread volatility predicts FP-beta dispersion, measured by $\beta_{\text{FP}}^H - \beta_{\text{FP}}^L$, the FP-predicted spread between the high and low beta portfolios underlying BAB.²² Consistent with Frazzini and Pedersen (2014), the prior month's realized TED-spread volatility is a negative predictor of FP-

²² This measure is closely related to Frazzini and Pedersen (2014) interquartile spread. Appendix F shows similar results using the standard deviation and mean absolute deviation of individual stocks' FP-beta estimates.

Table 4

Beta compression: predicting the beta spread.

The table reports results from time-series regressions predicting either 1) the spread in FP-predicted-betas between the high and low beta portfolios underlying BAB ($\beta_{FP}^H - \beta_{FP}^L$, Panel A), or 2) the one-month realized Dimson (1979) beta spread between these portfolios ($\beta_{1/12}^H - \beta_{1/12}^L$, Panel B). Explanatory variables are lagged one-month, and include the TED-spread volatility (σ_{TED} , measured as the standard deviation of daily changes to the TED-spread over the month), the level of the TED-spread (TED, measured as each month's latest available value as published by the Federal Reserve Bank of St. Louis), and the log-market volatility ratio, $\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$, as well as one month lags of the independent variables. Independent variables are demeaned and scaled by their sample standard deviations, so reported slope coefficients represent the predicted impact of a one standard deviation change in the explanatory variable at the mean. Reported t-statistics are calculated using Newey-West standard errors with 60 monthly lags. The sample covers January 1986 through December 2019, with the start date determined by the coverage of the St. Louis Fed's TED-spread data.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: $y = \beta_{FP}^H - \beta_{FP}^L$, spread in FP-betas used to leverage the portfolios underlying BAB						
Intercept	0.74 [19.27]	0.74 [19.43]	0.74 [20.89]	0.74 [35.17]	0.74 [36.42]	0.74 [35.09]
σ_{TED}	-0.02 [-1.14]		0.06 [1.99]		0.01 [0.75]	
TED		-0.04 [-3.93]	-0.09 [-3.22]			0.00 [0.10]
$\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$				-0.12 [-4.38]	-0.13 [-4.32]	-0.12 [-4.26]
adj.-R ² (%)	1.0	8.0	12.1	64.3	64.6	64.2
Panel B: $y = \beta_{1/12}^H - \beta_{1/12}^L$, spread in the portfolios' realized one-month betas						
Intercept	0.84 [15.13]	0.84 [15.15]	0.83 [13.61]	0.83 [14.13]	0.84 [28.29]	0.84 [30.62]
σ_{TED}	-0.05 [-2.61]					0.02 [0.54]
TED		-0.06 [-1.63]				-0.06 [-1.71]
$\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$			0.06 [1.93]			0.05 [1.49]
$\beta_{FP}^H - \beta_{FP}^L$				-0.05 [-1.64]		-0.00 [-0.18]
$\beta_{1/12}^H - \beta_{1/12}^L$					0.18 [5.51]	0.16 [4.52]
adj.-R ² (%)	2.4	2.8	3.3	2.1	26.8	28.5

beta dispersion, though it is not statistically significant and explains almost none of the time-series variation in this dispersion.²³ The second specification shows that the level of the TED-spread, which is highly correlated with the Frazzini and Pedersen (2014) measure of TED-spread volatility, is a significant negative predictor of FP-beta dispersion, and explains more of the time-series variation in this dispersion (8.0% versus 1.0%). The third specification shows that after controlling for the level of the TED-spread, TED-spread volatility is a significant *positive* predictor of FP-beta dispersion, a result that appears contrary to Frazzini and Pedersen (2014) prediction that “betas are compressed toward one when funding liquidity risk is high... [where] we use the volatility of the TED spread to

proxy for the volatility of margin requirements” (pg. 17). The fourth specification shows that the log-market volatility ratio, which explains 45% of the time-series variation in the FP-predicted market beta, explains even more of the time-series variation in the FP-beta dispersion, 64.3%. This is of course mechanical. Equation (2) for the FP-beta has the market volatility ratio in the denominator, practically guaranteeing that it is a strong, negative predictor. This of course does not prove that the log-market volatility ratio has any power predicting compression in actual betas. The log-market volatility ratio strongly predicts the value-weighted average bias in the FP-beta estimates, and this bias directly impacts the dispersion in the estimates. The fifth and sixth specifications show that neither the volatility or the level of the TED-spread have any marginal power predicting FP-beta dispersion controlling for the log-market volatility ratio.

Panel B shows strikingly different results using the spread in the portfolios' market betas actually realized over the following month as the dependent variable. The first specification shows that the TED-spread volatility is a marginally significant negative univariate predictor of the realized market beta spread, but explains only 2.4% of its time-series variation. The second specification shows that the level of the TED spread is not a market beta spread predictor. The third shows that the log-market volatility ratio, which explained 64.3% of the time-series variation in the predicted FP-beta spread, is not a significant predictor

²³ Interpretation of this result is further complicated, because Frazzini and Pedersen (2014) calculate TED-spread volatility using another non-standard procedure. Interest rate volatility is commonly measured as the annualized standard deviation of changes in the log-rate, but Frazzini and Pedersen (2014) measure TED-spread volatility each calendar month as the annualized standard deviation of daily changes in the level of the spread. TED-spread innovations tend to be proportional to the spread, so this measure of TED-spread volatility (or more accurately, TED-spread variability) is highly correlated with the TED-spread. The level of the TED-spread explains 65.9% of the variation in the TED-spread volatility measure employed by Frazzini and Pedersen (2014), despite explaining none of the variation (adj.-R² = 0.0%) in conventionally measured TED-spread volatility. For comparability we follow Frazzini and Pedersen (2014), using their unconventional measure of TED-spread volatility.

of the realized market beta spread between the high and low FP-beta portfolios. The fourth specification shows that the FP-predicted market beta spread between the high and low FP-beta portfolios has no power predicting the spread in these portfolios' future realized market betas. This is remarkable. Variation in the spread between the high and low beta portfolios' preformation betas is essentially unrelated to variation in the portfolios' realized post formation beta spread.

The last two specifications contrast this with the power that the lagged realized market beta spread has predicting the market beta spread. Market betas are persistent, so the portfolios' realized betas, even the noisy estimates obtained using a single month of daily returns, have significant power predicting the portfolios' betas in the following month. The lagged realized market beta spread is consequently a highly significant predictor, with a *t*-statistic exceeding six, explaining 26.8% of the time-series variation in the following month's realized market beta spread. The last specification shows that in a multiple regression the power of the past realized market beta spread to predict the market beta spread is almost undiminished, but that controlling for the past spread, none of the other variables have any power. Overall, Table 4 suggests that the evidence for beta compression that Frazzini and Pedersen (2014) present as support for their paper's underlying theory actually reflects bias in the paper's betas, driven by their non-standard beta estimation procedure. Neither funding constraints, nor even the FP-predicted beta spread between the high and low beta portfolios, have any power predicting dispersion in future realized market beta spreads.

The fact that the FP-predicted betas of the portfolios underlying BAB do not predict the spread between the portfolios' future realized market betas has strong implications for BAB's conditional performance. The portfolios' FP-predicted betas are both highly predictable and the basis of Frazzini and Pedersen (2014) hedging procedure. As a result, market volatility predicts the leverage BAB employs, but not the betas of the underlying portfolios, so BAB is mis-hedged in predictable ways. As a result, BAB is not conditionally market-neutral, and market volatility is a powerful predictor of BAB's market tilt.²⁴ This fact is further investigated in the next section.

5. BAB's beta

BAB dramatically over-weights nano- and micro-cap stocks (Section 3), and the FP-betas used to create and leverage BAB's underlying portfolios are biased in highly predictable ways (Section 4). Both of these facts raise concerns regarding whether BAB is truly market-neutral, as it is designed to be.

First, the small stocks disproportionately traded by BAB are prone to nonsynchronous trading problems.²⁵ The strategy is significantly net-long these nano- and micro-cap stocks, to the tune of 75 cents for each dollar invested in BAB. There is also a strong reason to suspect that the stocks held on the long side of the strategy are especially susceptible to this issue. These stocks are selected on the basis of having low estimated market betas, and nonsynchronous trading problems bias beta estimations downward.²⁶ Because BAB is significantly net-long nano- and micro-cap stocks, selecting especially for those with the largest nonsynchronous trading problems, we should expect that the strategy's market beta estimated at longer horizons exceeds its beta estimated at shorter horizons. Given that the strategy is designed to target a contemporaneous monthly market beta of zero, we should consequently expect the strategy to have a positive market tilt at longer horizons.

Second, given that the log-market volatility ratio explains 64.3% of the time-series variation in the FP-predicted beta spread between the high and low beta portfolios, and that these portfolios' FP-predicted betas are used to leverage these portfolios when constructing BAB, we should expect this ratio to also predict BAB's beta. When volatility is high, then the downward bias in the FP-betas should artificially compress the spread in the betas used to leverage the portfolios underlying BAB, resulting in under-hedging that is insufficient to fully offset the underlying low-minus-high beta strategy's short market tilt. Conversely, when volatility is low the bias should exaggerate the hedging beta spread, resulting in over-hedging and a long market tilt.

Table 5 tests both these sets of predictions, by performing a series of time-series regressions of the returns to BAB on the contemporaneous market return, lagged market returns, and market returns interacted with the log of the one-year to five-year market volatility ratio. The first specification shows that BAB is unconditionally close to contemporaneously market-neutral at the monthly frequency, with an average market loading of -0.06 that is barely significant (*t*-statistic of -1.92). The second specification includes one- and two-month lagged market returns as explanatory variables, and finds these to be much better predictors of BAB's performance. The loadings are larger, 0.16 and 0.12 respectively, and highly significant (*t*-statistics of 5.30 and 3.79), suggesting that BAB has a significant long market tilt at lower frequencies, a fact also reflected in the data—the market beta of BAB in annual data is 0.26 . This fact also helps explain the striking difference in betas observed on the daily and monthly BAB factors maintained by AQR. The monthly factor, similar to our replication, has a -0.07 contemporaneous beta in the post-1967 sample

²⁴ Frazzini and Pedersen (2014) also suggest that BAB's market tilt should be predictable, but for a very different reason. The FP-betas used to hedge BAB are highly predictable, but the market betas of the portfolios underlying BAB are not. Failing to understand the bias in their betas, Frazzini and Pedersen (2014) argue that the market betas of the portfolios underlying BAB can be better predicted using information about TED-spread volatility, which they argue is not incorporated into BAB's hedge.

²⁵ FP-betas are calculated using correlations estimated using three-day overlapping returns, a choice made explicitly to "control for nonsynchronous trading." The three day window is insufficient to resolve the problem, especially for the small stocks disproportionately traded by BAB.

²⁶ Baker et al. (2017) adopt the Frazzini and Pedersen (2014) procedure, calling the resulting estimates "improved measures of beta," despite seeming to recognize that doing so "has the effect of lowering the average betas of small stocks, which are individually less likely to trade in sync with the market overall, because of lower levels of liquidity" (p. 77).

Table 5

Predicting BAB's beta.

The table reports results from time-series regressions of the monthly returns to BAB and value-weighted beta-arbitrage on explanatory returns. Value-weighted beta-arbitrage buys the low beta third of stocks and sells the high beta third, where these betas are estimated each month using five years of daily data and Dimson corrected using one daily lag of market returns, and then hedges the resulting short market tilt by buying the market in proportion to the tilt observed over the preceding year. Explanatory variables are the contemporaneous market excess return (MKT), one- and two-month lagged market excess returns (MKT₋₁ and MKT₋₂), and the contemporaneous market excess return interacted with the one-month lagged market volatility ratio, $\ln(\sigma_1^{\text{MKT}}/\sigma_5^{\text{MKT}})$. The market volatility ratio is demeaned and scaled by its sample standard deviation, so the reported slope coefficients on $\text{MKT} \times \ln(\sigma_1^{\text{MKT}}/\sigma_5^{\text{MKT}})$ represent the predicted change in BAB's market beta resulting from a one standard deviation change in log market volatility ratio at the mean. The sample covers January 1968 through December 2019.

	$y = r_{\text{BAB}}$			$y = r_{\text{vw } \beta\text{-arb.}}$		
	(1)	(2)	(3)	(4)	(5)	(6)
$\text{MKT} \times \ln(\sigma_1^{\text{MKT}}/\sigma_5^{\text{MKT}})$			-0.13 [-4.61]			-0.05 [-1.79]
MKT	-0.06 [-1.93]	-0.07 [-2.25]	-0.02 [-0.53]	-0.02 [-0.88]	-0.02 [-0.71]	-0.00 [-0.04]
MKT ₋₁		0.16 [5.29]	0.18 [5.88]		-0.05 [-1.67]	-0.04 [-1.46]
MKT ₋₂		0.12 [3.80]	0.11 [3.57]		0.03 [0.98]	0.02 [0.88]
Intercept	1.11 [7.84]	0.97 [7.02]	0.94 [6.89]	0.59 [4.61]	0.59 [4.62]	0.58 [4.53]
adj.-R ² (%)	0.5	7.3	10.3	-0.0	0.2	0.6

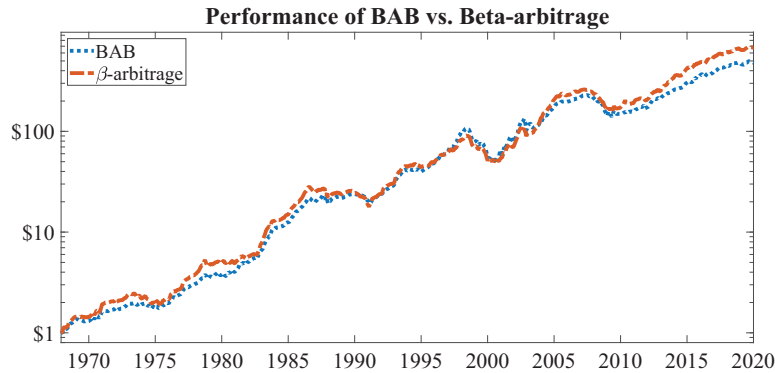


Fig. 9. BAB vs. simple equal-weighted beta-arbitrage. The figure shows the performance of our replication of BAB (dotted line), and simple equal-weighted beta-arbitrage, constructed using an equal-weighted tercile sort on directly estimated five-year stock betas, hedged by buying the equal-weighted market in proportion to the directly observed short-tilt of the underlying low-minus-high beta strategy (β -arbitrage, dot-dashed line). Because simple equal-weighted beta-arbitrage has a significantly lower volatility (9.5%), the figure shows the strategy leveraged to run at the same sample volatility as BAB (11.8%). The sample covers January 1968 through December 2019.

(t-statistic of -2.69). The magnitude of the short market tilt measured at the daily frequency, however, is significantly larger, -0.32 (t-statistic of -15.1 using Newey-West standard errors calculated with one month of daily lags), despite the fact that this daily factor's cumulated returns each month closely match the returns to the monthly factor. Understanding the magnitude of the nonsynchronous trading issues that plague BAB helps reconcile these seemingly disparate results.

The third specification explores BAB's conditional market tilt, by additionally including the contemporaneous market return interacted with the lagged market-volatility ratio as an explanatory return. The highly significant negative coefficient on the contemporaneous market return interacted with the lagged market volatility ratio is evidence that BAB, while unconditionally contemporaneously near market-neutral at the monthly frequency, tends to tilt away from the market when volatility is high, but towards

the market when volatility is low. That is, BAB's construction introduces market timing into the strategy, so that it takes less market risk when there is more uncertainty, similar to the "volatility managed" market strategy advocated by [Moreira and Muir \(2017\)](#).

Specifications four through six repeat these tests for the standard, implementable version of "beta-arbitrage." This strategy is based on the model of [Black \(1972\)](#). It is designed to be market-neutral, while taking a long position in Black's "beta factor," which [Black et al. \(1972\)](#) argue earns significant positive excess returns. The strategy is constructed using value-weighted portfolios formed on the basis of stocks' directly estimated five-year betas, and is much simpler than BAB. It buys the low beta portfolio, financing this position by shorting the high beta portfolio, and hedges the resulting short market position by directly buying the value-weighted market portfolio in proportion to the low-minus-high beta strategy's observed short mar-

ket tilt over the preceding year, financing this hedge by borrowing. The tests presented in Table 5 show that the nonsynchronous trading issues that plague BAB are absent from this beta-arbitrage. Beta-arbitrage is contemporaneously market-neutral (specification 4), and also market-neutral at longer horizons (specification 5). The last specification shows, consistent with the results of Cederburg and O'Doherty. (2016), that beta-arbitrage, like BAB, has a degree of volatility-related market timing, though the magnitude is less than half as large.

6. Conclusion

Frazzini and Pedersen (2014) Betting Against Beta (BAB) factor, based on the same basic idea as Black (1972) beta-arbitrage, exhibits striking backtested performance. This remarkable performance depends, however, on non-standard choices Frazzini and Pedersen (2014) make when constructing BAB. Their rank-weighting procedure is a backdoor to equal-weighting the strategy's underlying beta portfolios, and their leveraging procedure is a backdoor to hedging the underlying low-minus-high beta portfolio using these same portfolios. The resulting strategy, BAB, is little different from a simple, transparent version of beta-arbitrage, based on equal-weighted beta portfolios and hedged using the equal-weighted market. While this fact is obscured by Frazzini and Pedersen (2014) non-standard construction methods, it is readily apparent in Fig. 9, which shows the two strategies. The two are 84% correlated, and realize similar Sharpe ratios (1.07 for BAB and 1.12 for beta-arbitrage).

The strong paper performance of both strategies is driven by dramatically overweighting the market's smallest, least liquid stocks, while ignoring transaction costs and implementation issues. Accounting for transaction costs significantly reduces the strategies' profitability. While both strategies earn significantly positive average net returns, they earn these primarily by tilting towards profitability and investment, exposures for which they are

fairly compensated. Neither strategy has a significant net alpha relative to the Fama and French five-factor model.

The interpretation of BAB, and any empirical support for its underlying theory, is also severely compromised because Frazzini and Pedersen (2014) construct their factor using biased beta estimates. The paper's novel estimation procedure yields betas conflated with individual stock volatilities, which are biased in the time-series in ways strongly predictable by market volatility. Frazzini and Pedersen (2014) results on beta compression and conditional strategy performance reflect biases in their betas resulting from their non-standard estimation procedure, and thus cannot be taken as support for their underlying theory.

Overall, the results presented in BAB do not provide strong evidence for the profitability of defensive equity strategies.²⁷ Instead, they provide a powerful argument against the use of "sophisticated" or non-standard procedures. While novel techniques certainly have a role in economics, and are sometimes necessary to advance our knowledge, they are too often used to make results look stronger without yielding any deeper insights.

This provides a compelling rationale for "following the literature." No matter how arbitrary the choices made by the previous literature, they are past choices, and following them now reduces the degrees of freedom available for overfitting the data when doing empirical research (see also Novy-Marx, 2018). As a consequence, deviations from well-accepted methods should be both well motivated and well understood. When results are obtained using non-standard methods, the first question should be "What are the results using standard methods?" If the answer is "different, but those obtained using non-standard methods are still interesting," the next question should be "Why? And what, exactly, is driving any differences?" When a non-standard method's impact on results is not obvious, or the mechanism driving any difference is not clear, its real impact may well be different than what is claimed.

²⁷ Novy-Marx (2016) also shows that the performance of "defensive" strategies that bet against volatility can be understood by the tilts they take to profitability and value, at least after controlling for size.

Appendix A. Comparison of BAB factors

Fig. A.10 shows the cumulative performance of the three versions of BAB: the original series used in Frazzini and Pedersen (2014) paper, the BAB currently maintained by AQR, and our replication.

Table A.6 shows that both our replication and the original BAB factor have highly significant alphas relative to the currently maintained factor (specifications 5 and 8), while the converses are false (specifications 2 and 3). It also shows that our replication and the original factor do not have significant alphas relative to each other (specifications 6 and 9).

Appendix B. Rank-weighted vs equal-weighted value and momentum

Fig. B.11 shows the nearly identical performance of rank-weighted and equal-weighted (top-minus-bottom third) strategies, both for value and momentum. For the value strategies, based on book-to-market, the two methods yield statistically indistinguishable average returns, and a monthly return correlation of 99.7%. The correlation between the two momentum strategies is even higher, at 99.8%.

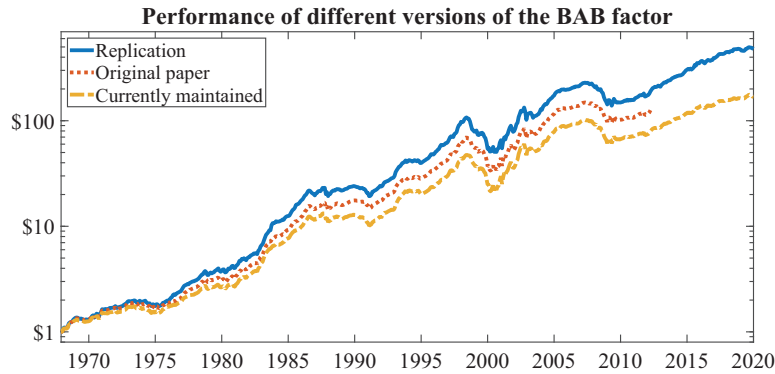


Fig. A.10. Performance comparison of different versions of the BAB factor. The figure shows the performance of three different versions of the Frazzini and Pedersen (2014) Betting-Against-Beta factor. The top, solid line shows our replication of the factor; the middle, dotted line shows the BAB data used in the original paper, which ends after March 2012; and the bottom, dot-dashed line shows the factor currently maintained by AQR. The latter two series are available at <https://www.aqr.com/Insights/Datasets/>. The sample covers January 1968 through December 2019.

Table A.6

BAB strategy performances, three different versions of BAB The table reports results from time-series regressions of the form

$$y = \alpha + \beta'X + \epsilon,$$

using the monthly returns to three different versions of BAB: the equity factor currently maintained by AQR (BAB), the factor studied in Frazzini and Pedersen (2014) (BAB-orig.), and our replication of the factor (BAB-rep.). The sample covers January 1968 through December 2019.

	$y = r_{BAB}$			$y = r_{BAB-orig.}$			$y = r_{BAB-rep.}$		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
α	0.88 [6.58]	-0.12 [-3.26]	-0.10 [-2.63]	0.97 [6.70]	0.15 [4.65]	0.02 [0.53]	1.05 [7.69]	0.19 [4.67]	0.03 [0.77]
r_{BAB}					0.92 [101.09]			0.98 [84.06]	
$r_{BAB-orig.}$		1.04 [101.09]							1.06 [112.31]
$r_{BAB-rep.}$			0.94 [84.06]			0.90 [112.31]			
adj.- R^2 (%)		95.1	91.9		95.1	96.0		91.9	96.0

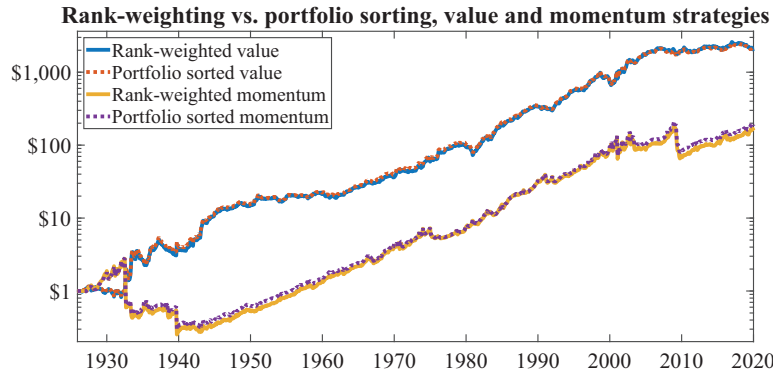


Fig. B.11. Rank-weighting vs. portfolio sorting, value and momentum strategies. This figure compares the performance of strategies based on rank-weighted portfolios (solid lines), and from a simple sort into equal-weighted portfolios holding the top and bottom third of stocks (dotted lines). The top two lines show value strategies, sorted on the basis of book-to-market and rebalanced annually at the end of June. The bottom two lines show momentum strategies, sorted on the basis of stock performance realized over the first eleven months of the preceding year and rebalanced at the end of each month. The sample covers January 1927 through December 2019.

Appendix C. Performance details of BAB variations

Table C.7 shows results of time-series regressions using our replication of BAB, and the simple version of beta-arbitrage (β -arb.) to which it is most similar, discussed in this paper's Conclusion. This beta-arbitrage strategy is based on an underlying low-minus-high beta strategy, constructed using an equal-weighted tercile sort (top and bottom 33.3%) on betas estimated directly from CAPM regressions using the preceding five years of daily returns, and hedged by buying the equal-weighted market in proportion to the underlying strategy's beta estimated directly from its realized returns.

The table shows that simple beta-arbitrage has a slightly higher Sharpe ratio than BAB (specifications 1 and 6), and similar alphas relative to the standard three-, five-, and six-factor models. The two strategies are highly correlated (84%), and have small, marginally significant alphas relative to each other.

Table C.8 shows results of time-series regressions using the BAB-like strategies based on underlying beta portfolios constructed from simple tercile sorts, considered in Section 2.1. One version, "equal-weighted BAB" (EWBAB) is based on equal-weighted portfolios that hold the top and bottom thirds of stocks by estimated beta. The other, VW-

Table C.7

Performance of BAB and simple equal-weighted beta-arbitrage. The table reports results from time-series regressions of the form

$$y = \alpha + \beta'X + \epsilon,$$

using the monthly returns to our replication of BAB (BAB), and the simple version of beta-arbitrage to which it is most similar (β -arb.). This is based on an underlying low-minus-high beta strategy, constructed using an equal-weighted tercile sort (top and bottom 33.3%) on betas estimated directly from CAPM regressions using the preceding five years of daily returns, and hedged by buying the equal-weighted market in proportion to the underlying strategy's beta estimated directly from its realized returns. The sample covers January 1968 through December 2019.

	$y = r_{BAB}$					$y = r_{\beta - arb.}$				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
α	1.05 [7.69]	0.94 [7.05]	0.65 [5.11]	0.51 [4.13]	0.12 [1.53]	0.89 [8.09]	0.72 [6.97]	0.60 [5.73]	0.54 [5.09]	0.18 [2.86]
MKT		-0.01 [-0.23]	0.07 [2.29]	0.10 [3.33]			0.07 [2.97]	0.11 [4.28]	0.12 [4.85]	
SMB		0.03 [0.75]	0.17 [3.89]	0.17 [3.98]			0.18 [5.24]	0.22 [6.09]	0.22 [6.13]	
HML		0.37 [7.88]	0.18 [3.02]	0.30 [5.03]			0.34 [9.26]	0.21 [4.15]	0.26 [5.19]	
RMW			0.56 [9.25]	0.52 [8.80]				0.19 [3.81]	0.17 [3.39]	
CMA			0.39 [4.33]	0.30 [3.42]				0.24 [3.23]	0.20 [2.64]	
UMD				0.21 [7.22]					0.10 [4.15]	
β -arb.					1.05 [39.1]					
BAB										0.68 [39.1]
adj.- R^2 (%)		9.3	21.0	27.0	71.0		14.4	16.8	19.0	71.0

Table C.8

Performance of equal-weighted BAB and value-weighted BAB. The table reports results from time-series regressions of the form

$$y = \alpha + \beta'X + \epsilon,$$

using the monthly returns to the BAB-like strategies based on underlying beta portfolios constructed from simple tercile sorts, considered in [Section 2.1](#). One version, “equal-weighted BAB” (EWBAB) is based on equal-weighted portfolios that hold the top and bottom thirds of stocks by estimated beta. The other, “value-weighted BAB” (VWBAB), is based on value-weighted portfolios that hold the same stocks. The sample covers January 1968 through December 2019.

	$y = r_{EWBAB}$				$y = r_{VWBAB}$				$y = r_{BAB}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
α	1.02 [7.53]	0.62 [4.94]	0.49 [3.95]	−0.02 [−1.49]	0.54 [3.47]	0.24 [1.68]	0.14 [0.96]	−0.21 [−1.59]	0.03 [2.11]	0.76 [6.99]
MKT		0.07 [2.14]	0.09 [3.18]			−0.06 [−1.60]	−0.03 [−0.99]			
SMB		0.17 [3.83]	0.16 [3.92]			−0.05 [−1.02]	−0.05 [−1.08]			
HML		0.18 [3.02]	0.30 [5.05]			0.16 [2.32]	0.25 [3.56]			
RMW		0.56 [9.40]	0.52 [8.95]			0.44 [6.44]	0.41 [6.02]			
CMA		0.40 [4.48]	0.31 [3.56]			0.55 [5.32]	0.48 [4.67]			
UMD			0.21 [7.29]				0.16 [4.66]			
BAB				0.99 [293.1]				0.71 [19.8]		
EWBAB									1.00 [293.1]	
VWBAB										0.54 [19.8]
adj.-R ² (%)		21.7	27.8	99.3		23.8	26.3	38.6	99.3	38.6

BAB, is based on value-weighted portfolios that hold the same stocks.

The table shows that BAB and equal-weighted BAB are basically indistinguishable. They are 99.6% correlated, and the magnitudes of their alphas relative to the other are less than 3 bps/month (specifications 4 and 9). Value-weighted BAB is far less similar to these strategies, has a Sharpe ratio less than half as high (specification 5), and does not have a significant alpha relative to the five- or six-factor models (specifications 6 and 7), or to equal-weighted BAB (specification 8).

[Table C.9](#) shows results of time-series regressions using the returns to the “directly hedged BABs” (DHBABs) considered in [Section 2.2](#). These are based on the dollar-neutral rank-weighted low-minus-high beta strategy underlying BAB, but hedged by directly buying the market instead of using [Frazzini and Pedersen \(2014\)](#) leveraging procedure. One version is hedged by buying the equal-weighted market in proportion to the underlying strategy’s negative market tilt estimated using its prior year of daily realized returns (DHBAB-EW). The other is constructed the same way but hedged using the value-weighted market (DHBAB-VW).²⁸

The table shows that BAB directly hedged using the equal-weighted market has a higher Sharpe ratio than BAB (specification 1). It is also more than 90% correlated with

BAB, relative to which it has large, highly significant alpha (specification 4), but which it prices well (specification 9). Specifications 5–7 show that the version directly hedged by buying the value-weighted market is less profitable though it still has highly significant alphas relative to the standard models. It is also less similar to BAB (correlation of 72%), but priced by BAB (specification 8).

Finally, [Table C.10](#) shows the performance, net of transaction costs, of the BAB-like strategies considered in [Section 3.1](#). The three strategies are BAB; a rank- and cap-weighted BAB based on the procedure discussed in [Footnote 5](#), which holds stocks with a size distribution more reflective of that observed in the market (BAB^{cap}); and a version of BAB constructed just like BAB but in the universe of stocks that excludes the smallest NYSE size decile (BAB^{ex}).

The table shows the three strategies’ average net returns, and their generalized alphas (Novy-Marx and Velikov, 2016) relative to the market model, and to the Fama and French three- and five-factor models. The generalized alpha measures the extent to which a test asset could have improved the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the test asset and the explanatory factors. While the generalized alpha agrees exactly with the common notion of alpha when trading is frictionless, when trading is costly it cannot be calculated by simply projecting the test asset’s net returns onto the factors’ net returns, which makes the test asset’s factor loadings uninformative. Consequently, to provide a sense of how the test assets improve the investment opportunity

²⁸ Alternatively, these “directly hedged BABs” can be thought of as versions of beta-arbitrage based on rank-weighted underlying beta portfolios (rank-weighted on the FP-betas).

Table C.9

Performance of directly hedged BAB. The table reports results from time-series regressions of the form

$$y = \alpha + \beta'X + \epsilon,$$

using the monthly returns to “directly hedged BABs” considered in Section 2.2. These are based on the dollar-neutral rank-weighted low-minus-high beta strategy underlying BAB, but hedged by directly buying the market instead of using Frazzini and Pedersen (2014) leveraging procedure. One version, BAB_{EW}^{DH} , is hedged by buying the equal-weighted market in proportion to the underlying strategy's negative market tilt estimated using its prior year of daily realized returns. The other, BAB_{VW}^{DH} , is constructed the same way but hedged using the value-weighted market. The sample covers January 1968 through December 2019.

	$y = r_{DHBAB-EW}$				$y = r_{DHBAB-VW}$				$y = r_{BAB}$	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
α	0.99 [8.99]	0.71 [6.51]	0.62 [5.75]	0.23 [4.56]	0.80 [5.85]	0.57 [4.95]	0.40 [3.66]	0.05 [0.48]	−0.06 [−0.93]	0.48 [4.87]
MKT		0.13 [4.78]	0.15 [5.55]			0.16 [5.57]	0.19 [7.43]			
SMB		0.17 [4.39]	0.16 [4.43]			−0.50 [−12.4]	−0.50 [−13.6]			
HML		0.09 [1.82]	0.17 [3.25]			0.04 [0.78]	0.20 [3.79]			
RMW		0.26 [5.00]	0.23 [4.51]			0.45 [8.22]	0.40 [7.75]			
CMA		0.28 [3.59]	0.22 [2.87]			0.28 [3.43]	0.16 [2.14]			
UMD			0.13 [5.28]				0.27 [10.7]			
BAB				0.73 [52.1]				0.72 [25.6]		
BAB_{EW}^{DH}									1.12 [52.1]	
BAB_{VW}^{DH}										0.71 [25.6]
adj.- R^2 (%)		11.3	15.0	81.3		35.3	45.3	51.2	81.3	51.2

set relative to each factor model, the table reports ex post mean variance efficient tangency portfolio weights and the maximum attainable Sharpe ratios, accounting for transaction costs.

Panel A shows that the three strategies realized remarkably similar average net returns, 42–47 bps/month, and similar Sharpe ratios. They did so despite having highly disparate costs of trading, ranging from only 8 bps/month for rank- and cap-weighted BAB to almost 60 bps/month for BAB (over the last ten years of the sample these costs were only 4 and 17 bps/month, respectively; the cost of trading BAB ex-nanos lies in between, at 22 bps/month over the whole sample, and 9 bps/month over the late sample). The strategies' generalized alphas relative to the five-factor model (Panel D) are similarly close, ranging from 15 to 19 bps/month, with t-statistics ranging from 1.31 to 1.40. An investor already trading the five Fama and French factors could not in practice have realized significant gains by additionally trading any of these versions of BAB (maximal Sharpe ratio improvements from 0.90 to 0.92), and the ex post efficient portfolio of the factors plus any of the versions of BAB takes only a modest position in BAB (7–8% of the portfolio).²⁹

²⁹ Panel C shows that all three versions of BAB are correlated with value (31–41%), observable in the weight reduction on HML in the ex post efficient portfolio that results from adding BAB to the investment opportunity set (reductions from 0.54 down to 0.24–0.32), but that all three BABs have significant generalized alphas relative to the three-factor model (38–40 bps/month with t-statistics ranging from 2.79 to 3.13). In Panel D CMA pushes HML out of the ex post efficient portfolios. CMA is a good sub-

Appendix D. Non-standard FP-betas' impact on BAB performance

In addition to the time-series bias, using FP-betas instead of market betas has a second potential effect on the average performance of BAB. In the cross-section, the scaling of the FP-betas by the market volatility ratio in Eq. (2) is unimportant, because it is a multiplier on all the estimates. Sorting on the FP-beta is, however, equivalent to sorting on the product of 1) stocks' directly estimated five-year betas, β_5^i , and 2) the ratio of individual stocks' volatilities estimated at one- and five-year horizons, σ_1^i/σ_5^i . Overall this suggests two potential effects of the FP-beta estimation procedure on BAB's average performance, one driven by stock selection, because cross-sectionally stocks are ranked on the basis of a combination of their five-year betas and their current volatilities relative to their longer term means, $\beta_5^i\sigma_1^i/\sigma_5^i$, and one driven by the leveraging procedure, because the FP-beta estimates have a procyclical bias in the time-series, at least to the extent that market volatility is countercyclical.

Fig. D.12 shows that the net impact of the two effects on BAB's average performance is negligible. In particular, it shows that an “almost BAB” strategy, constructed like BAB but which rank-weights and leverages the underlying

stitute for HML, because they are highly correlated (71%) and have similar Sharpe ratios, and investment is more complementary to operating profitability than is value: the correlation between CMA and RMW is 1%, while the correlation between HML and RMW is 9%.

Table C.10

BAB performance net of transaction costs. The table reports the performance of BAB net of transaction costs using the generalized alpha of Novy-Marx and Velikov (2016), which measures the extent to which adding a test asset improves the investment opportunity set relative to a set of explanatory factors accounting for the cost of trading. In addition to BAB, it reports results for rank- and capitalization-weighted BAB (BAB^{cap}), and for BAB constructed in a universe that excludes the nano-cap stocks from the smallest NYSE size decile (BAB^{ex}). The table also reports the weight of each asset in the achievable (net of transaction costs) ex post MVE portfolio. The sample covers January 1968 through December 2019.

Factor weight in ex post MVE portfolio									
α^*	Test asset			Explanatory factors					SR
	BAB	BAB ^{cap}	BAB ^{ex}	MKT	SMB	HML	RMW	CMA	
Panel A: No model (average net returns)									
0.47	1.00								0.46
[3.35]									
0.42		1.00							0.39
[2.82]									
0.46			1.00						0.47
[3.39]									
Panel B: One-factor model				1.00					0.41
0.50	0.59			0.41					0.65
[3.59]									
0.50		0.54		0.46					0.66
[3.74]									
0.54			0.59	0.41					0.69
[4.00]									
Panel C: Three-factor model				0.39	0.06	0.54			0.60
0.38	0.33			0.32	0.04	0.32			0.71
[2.79]									
0.40		0.31		0.32	0.10	0.27			0.73
[2.95]									
0.39			0.36	0.30	0.10	0.24			0.74
[3.13]									
Panel D: Five-factor model				0.20	0.09	0.00	0.32	0.40	0.90
0.17	0.07			0.20	0.08	0.00	0.28	0.37	0.92
[1.32]									
0.19		0.07		0.20	0.09	0.00	0.28	0.35	0.92
[1.40]									
0.15			0.08	0.20	0.09	0.00	0.27	0.36	0.92
[1.31]									

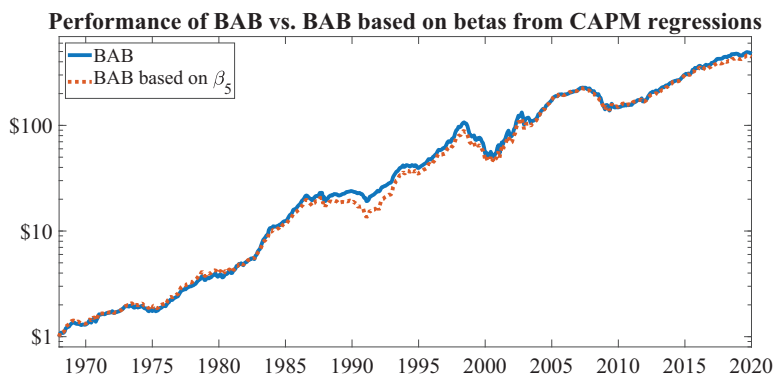


Fig. D.12. BAB vs. BAB based on betas from CAPM regressions. The figure shows the performance of our replication of BAB (dotted line), and an “almost BAB” factor (solid line), built using the same methods used to construct BAB, but based on individual stock betas directly estimated from CAPM regressions using five years of overlapping three-day returns, as opposed to the FP-betas, which are constructed by combining correlations estimated using five years of overlapping three-day returns and volatilities estimated using one year of daily data. The sample covers January 1968 through December 2019.

portfolios on the basis of directly estimated five year betas instead of the FP-betas, performs almost indistinguishably from BAB. The two strategies have monthly return series that are 94.7% correlated, and realize almost the same Sharpe ratio over the sample period (1.07 vs. 1.05). There is also no advantage, from an investment perspective, to trading either strategy; neither has a significant alpha relative to the other, either alone or including the returns to

the most common asset pricing factors (Fama-French five plus UMD) as explanatory returns.

Untabulated results show that BAB has a small but significant alpha (10 bps/month with a t-statistic of 2.30) relative to a BAB-like strategy based on beta portfolios rank-weighted on the basis of the directly estimated five-year betas but leveraged each month exactly like BAB. Conversely, a BAB-like strategy based on the same underlying

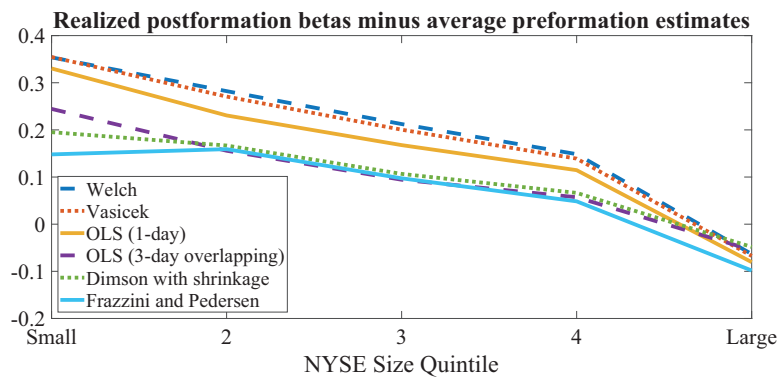


Fig. E.13. Postformation-minus-preformation beta spreads by size quintile. The figure shows the extent by which the realized betas of the NYSE size quintiles exceed their time-series average preformation estimates. Realized portfolio betas are calculated over the full sample using CAPM regressions and monthly returns. Preformation estimates are calculated as the weighted average of the beta estimates of the individual stocks each portfolio holds at the start of each month. Preformation estimates are made using six different beta estimators: OLS estimates calculated using either one year, of daily returns or one year of overlapping three-day returns; the estimators of Vasicek (1973) and Welch (2019); the FP-beta estimator; and Dimson corrected estimates calculated using one year of daily returns that employ one lag and the same degree of shrinkage used by Frazzini and Pedersen (2014). The sample covers January 1968 through December 2019.

ing beta portfolios as BAB, but leveraged like the BAB-like strategy based on the directly estimated five-year betas, has a small, significant alpha relative to BAB (6 bps/month with a t-statistic of 2.52). That is, it looks as if sorting on the basis of the FP-betas yields marginal improvements to BAB's back-tested performance, but that leveraging on the basis of the FP-betas yields offsetting performance reductions.

Appendix E. Bias in beta estimates

Common beta estimates made using daily data tend to be downward biased because of the nonsynchronous trading problem. The average magnitude of this bias is negatively correlated with firm size, because the nonsynchronous trading problem is more acute for smaller firms. This fact is apparent looking at the extent to which size-sorted portfolios' realized betas exceed their preformation estimates. Fig. E.13 shows this for size quintiles. Realized betas are the portfolios' market loadings over the full sample (January 1968 through December 2019) estimated from CAPM regressions performed using monthly data. Pre-formation betas are the time-series averages of the value-weighted betas of the portfolios' holdings at the beginning of each month, where individual stock betas are estimated using one of six procedures: simple OLS with one year of either daily returns or three-day overlapping returns, the improved estimates of Dimson (1979), Welch (2019), and Vasicek (1973), also estimated using one year of daily data, and the split-frequency estimator of Frazzini and Pedersen (2014), which calculates volatilities using one year of daily data and correlations using five years of overlapping three-day returns. The figure shows that the realized beta for the largest 20% of firms is actually slightly lower than the average preformation estimates for any of the six beta estimators. As one moves down the size distribution, however, the portfolios' realized betas become large relative to preformation estimates. The bias is particularly acute for the estimators that make no attempt

to control for the asynchronous trading problem (OLS using only contemporaneous daily returns, and the estimators of Welch and Vasicek), but is obviously apparent even for the estimators that do (OLS using three-day overlapping daily returns, and the estimators of Dimson and FP). Spreads are on average lower for the FP-betas, because these betas have, on average, a positive value-weighted average bias. That is, the fact that on average the market predicted FP-beta is higher than one (biased upward) partly offsets the downward bias in the small stocks resulting from the nonsynchronous trading, yielding the smaller spreads between realized portfolio betas and average preformation estimates.

E1. Increasing beta spread by adjusting estimates for market capitalization

The fact that the nonsynchronous trading problem is stronger in the small cap universe drives an attenuation bias when sorting stocks on the basis of estimated betas. Small stocks with beta estimates just too low for inclusion in the high beta portfolio have, on average, higher true betas than large cap stocks with beta estimates just high enough for inclusion in the high beta portfolio. This reduces the beta spread realized between the high and low estimated beta portfolios. We should expect, therefore, that correcting for the nonsynchronous trading bias helps generate a greater realized beta spread in beta sorted portfolios. We show that this is indeed the case. We do not attempt to correct for the bias in any optimal way, which is beyond the scope of this paper, but simply show that even the simple correction of adjusting beta estimates by the percentile rank of firms' market capitalizations at the time of portfolio formation yields significant increases in the realized postformation beta spreads for all the beta estimators considered in this paper.

Fig. E.14 shows full-sample market loadings, estimated using monthly returns from January 1968 through December 2019, for high-minus-low decile portfolios sorted on

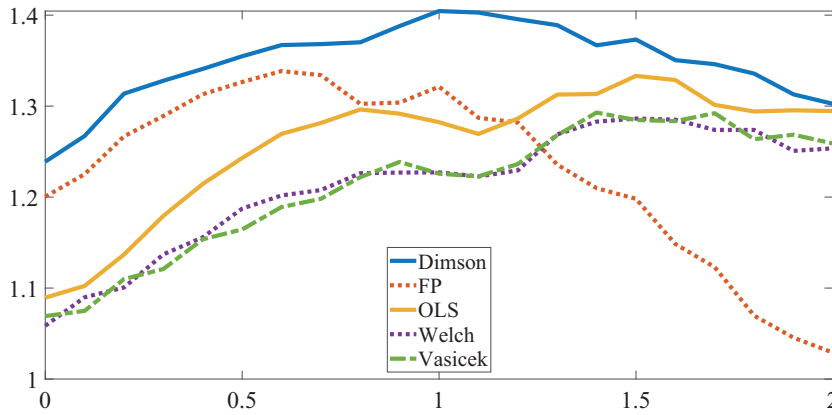


Fig. E.14. Realized beta spreads from estimators corrected for firm market capitalization. The figure shows the spread in betas between the high and low deciles realized over the full sample, January 1968 through December 2019. Portfolios are formed by sorting each month on size-adjusted beta estimators. Five estimators are used, a simple OLS estimate calculated using one year of daily returns; the estimators of Vasicek (1973) and Welch (2019); the FP-beta estimator; and the OLS estimate Dimson-corrected using one lag in daily returns. Estimates are corrected cross-sectionally by subtracting off the firms' market capitalization percentile rank ($rank_{ME}$), weighted by a factor between zero and two (w , the weight on $-rank_{ME}$, the x-axis).

beta estimators simply adjusted for size. Specifically, these portfolios are constructed by sorting on

$$\hat{\beta} - w \cdot rank_{ME},$$

where $\hat{\beta}$ is one of five different beta estimators (simple OLS using one year of daily data, the Dimson estimator corrected using one daily lag, the estimators of Vasicek and Welch, and the FP-beta estimator), $rank_{ME}$ is the percentile ranking of firms' market equity, and w is the weight put on the size adjustment. The left-hand side of the figure (spreads when there is no weight on market cap, i.e., from sorting on beta estimators alone) agrees with the right-hand side of Fig. 5 (which shows realized beta spreads for sorts on different beta estimators into different numbers of portfolios). Most strikingly, the figure shows that for all five estimators, adjusting beta estimates upward for small cap stocks relative to large cap stocks increases the realized beta spread. Adjusting beta estimates for market capitalization yields the greatest improvement for estimators more afflicted by the nonsynchronous trading problem, i.e., those that use only contemporaneous daily returns: simple OLS, and the estimator of Dimson and Welch. These estimators also improve more with larger adjustments (bigger w).

E2. Controlling for one beta estimator when sorting on another

To investigate the extent to which different beta estimators contain similar (or dissimilar) information, we perform a series of conditional sorts. These sorts control for one variable, by selecting small groups of stocks almost identical on the control characteristic (i.e., propensity-matched on the control variable), and then from each group assign the most dissimilar on the basis of the sorting variable to high and low portfolios. Essentially this is a very fine conditional double sort, where all the high portfolios from the secondary sort are aggregated into a single overall high portfolio, and similarly on the low side.

For comparability we would like this procedure to yield a similar spread in portfolio weighted-average sorting variable ranks as that observed from the Frazzini and Pedersen (2014) sorting procedure. Earlier we saw that simply sorting into three portfolios yields the same spread in weighted-average sorting variable rank as the Frazzini and Pedersen (2014) procedure of signal-weighting portfolios, a weighted average rank-spread of two-thirds. We use this to guide the exact construction of our conditionally sorted portfolios.

Given n randomly selected stocks, the expected spread in the extreme order statistics of the stocks' full cross-sectional ranks on any underlying characteristic is

$$E[U_{(n,n)} - U_{(1,n)}] = 1 - 2E[U_{(1,n)}] = \frac{n-1}{n+1},$$

where $U_{(k,n)}$ denotes the k th highest order statistic from n draws from a standard uniform distribution. The first equality follows from symmetry and the second from the fact that $U_{(k,n)} \sim \text{Beta}(k, n+1-k)$.

Sorting into N equal size portfolios on the basis of any characteristic yields a high-minus-low spread in sorting variable rank of $1 - 1/N$, as shown in Section 2.1. Taken together, this implies that the expected rank spread in extreme order statistics from $n = 2N - 1$ randomly selected stocks is the same as the high-minus-low spread from a sort into N portfolios on the basis of the characteristic.

Our empirical work consequently focuses on portfolios constructed from groups of five stocks propensity-matched on the control characteristic. Selecting groups of five stocks at random (i.e., using a placebo control variable) and assigning those with the extreme order statistics of the sorting characteristic to the high and low portfolios yields the same expected spread in sorting variable rank, $\frac{5-1}{5+1} = 2/3$, as for portfolios rank-weighted directly on the sorting characteristic.

Table E.11 shows full sample market loadings, estimated from CAPM regressions using monthly returns, for stocks sorted on betas estimated using five different procedures

Table E.11

Conditional beta spreads The table reports full sample market loadings from monthly CAPM regressions using the returns to strategies formed by sorting on one beta estimator after controlling for another. In particular, strategies are high-minus-low portfolios, formed by first propensity matching stocks into groups of five on the basis of the control variable (rows), and then assigning the extremes from each group of five on the basis of the sorting variable (columns) to the high and low portfolios. The beta estimators considered include a CAPM estimate obtained using contemporaneous daily market returns (OLS), the estimators of Vasicek, Welch, and FP, and an OLS estimate Dimson-corrected by including the loading on one day lagged market returns. All the betas are estimated monthly, using one year of daily returns, except for the FP-beta estimator which requires five years of daily returns. We also consider market equity (ME) as both a sorting variable and a control variable. For comparison, we also present results using a random, “placebo” control variable (“Noise”), as well as results from simple tertile sort on the basis of the stock selection variable (i.e., no control, labeled “None”, where three portfolios are chosen here to generate spread similar to that obtained by taking the extreme stocks from groups of five stocks selected at random, i.e., the placebo control). Portfolios are rebalanced monthly, and returns are value weighted. The sample covers January 1968 through December 2019.

Variable used for propensity-score matching	Variable used to select stocks for the high and low portfolios					
	$\hat{\beta}_{OLS}$	$\hat{\beta}_{Vas.}$	$\hat{\beta}_{SW}$	$\hat{\beta}_{FP}$	$\hat{\beta}_{Dim.}$	ME
None	0.69	0.67	0.68	0.75	0.79	−0.08
Noise	0.66	0.67	0.68	0.69	0.71	−0.17
$\hat{\beta}_{OLS}$		0.42	0.32	0.31	0.33	−0.42
$\hat{\beta}_{Vas.}$	−0.38		0.14	0.26	0.23	−0.50
$\hat{\beta}_{Welch}$	−0.19	−0.03		0.25	0.24	−0.50
$\hat{\beta}_{FP}$	0.14	0.21	0.22		0.26	−0.30
$\hat{\beta}_{Dim.}$	−0.03	0.07	0.12	0.20		−0.33
ME	0.75	0.75	0.76	0.73	0.74	

(simple OLS, Vasicek, Welch, FP, and Dimson) and market equity (ME), from groups of five stocks propensity-matched on the same variables. For reference, the first row shows full sample betas from simple sorts into tertile portfolios on the basis of each of the six variables, and is labeled “None” (i.e., no conditioning variable). Consistent with the spreads observed in Fig. 5, we see tertile spreads on the order of 0.7, with the Dimson estimates yielding the largest spread (0.79), the FP-beta estimates doing almost as well (0.75), and the three estimators based solely on contemporaneous daily returns faring slightly worse (spreads from 0.67–0.69). Sorting on market cap yields a modest negative beta; on average large cap stocks have slightly lower market betas than large cap stocks.

The second row shows results from the propensity-matched sorting procedure using a placebo control variable, labeled “Noise” (again, the conditioning variable), and justifies the theory behind using propensity-matched groups of five stocks. The observed spreads resulting from sorting into five portfolios using the placebo conditioning are similar to those obtained by simply sorting into three portfolios.

Rows three through seven show results of conditioning on each of the beta estimators when sorting on the others. The third row shows results for sorts propensity-matched on the OLS beta estimator, labeled $\hat{\beta}_{OLS}$. In every case, the realized full sample market loading that results from sorting on the other beta estimates after controlling for the OLS estimates remains large, from 0.31–0.42, roughly half as large as with the random control. At the same time, the first column shows that sorting on the OLS estimates after controlling for the others yields much smaller spreads: a modest 0.14 when controlling for FP-beta, close to zero when controlling for the Dimson estimates, and negative loadings when controlling for the Vasicek and Welch estimates. In fact, the magnitude of the spread between those stocks with the highest and lowest OLS beta estimates from among stocks with nearly identical Vasicek estimates is more than half as large as the spread obtained using random conditioning, but negative.

That is, after conditioning on the Vasicek estimate, a high OLS estimate is a strong predictor of a low beta going forward.

The fourth and fifth rows of Table E.11 show that the Vasicek and Welch estimators provide little incremental information relative to each other, in the sense that sorting on one after controlling for the other yields little realized beta spread, while the FP and Dimson estimators still yield significant but modest spreads after controlling for the Vasicek and Welch estimators ranging from 0.23–0.26. The sixth row shows that the FP-beta estimator yields some information relative to all the others, generating conditional realized spreads of 0.14–0.26 after controlling for the others. This should be expected, as the FP-beta estimator is calculated using returns over the preceding five years, while all the others are estimated using only one year of daily data. The seventh row shows that controlling for the Dimson estimates yields, on average, the smallest conditional spreads, while the fifth column shows that the spreads generated by sorting on the Dimson estimates after controlling for the other estimators are always larger than the spreads one obtains sorting on the other estimators after controlling for the Dimson estimates.

Finally, the last row shows results from conditioning on market capitalization. Consistent with Fig. E.14, controlling for size increases the spreads observed from sorting on all five beta estimators relative to those realized from random conditioning. The increase is larger for those estimators most prone to the bias resulting from the non-synchronous trading problem, simple OLS and the estimators of Vasicek and Welch. The last column shows remarkably large spreads from sorting on market capitalization after propensity matching on the beta estimates, especially when the control is based on one of these same three estimators. That is, among stocks with similar beta estimates, small cap stocks have, on average, much larger true betas. In the case of the Vasicek and Welch estimators, the magnitude of the realized spread obtained by sorting on market capitalization among stocks with nearly identical beta estimates is three-quarters as large as the spread obtained

Table F1

Beta compression: predicting alternative spread measures The table reports results from time-series regressions predicting either 1) the cross-sectional standard deviation in individual stocks' estimated FP-beta ($\sigma_{\beta_{FP}}$, Panel A), or 2) the mean absolute deviation of these betas ($|\beta_{FP} - \bar{\beta}_{FP}|$, Panel B). Explanatory variables are one-month lagged z-scores for the TED-spread volatility (σ_{TED} , measured as the standard deviation of the daily innovations to the TED-spread over the month), the level of the TED-spread (TED, measured as each month's latest available value as published by the Federal Reserve Bank of St. Louis), and the log-market volatility ratio ($\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$). Reported t-statistics are calculated using Newey-West standard errors with 60 monthly lags. The sample covers January 1986 through December 2019, with the start date determined by the coverage of the St. Louis Fed's TED-spread data.

	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: $y = \sigma_{\beta_{FP}}$, standard deviation of individual stocks' FP-betas estimates						
Intercept	0.34 [19.67]	0.34 [20.05]	0.34 [21.73]	0.34 [36.05]	0.34 [36.55]	0.34 [35.33]
$z_{\sigma_{TED}}$	-0.01 [-1.29]		0.03 [2.18]		0.00 [0.38]	
z_{TED}		-0.02 [-4.19]	-0.04 [-3.79]			-0.00 [-0.66]
$\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$				-0.05 [-4.22]	-0.05 [-4.10]	-0.05 [-3.95]
adj.-R ² (%)	1.5	9.7	14.0	57.7	57.6	57.7
Panel B: $y = \beta_{FP} - \bar{\beta}_{FP} $, mean absolute deviation of stocks' FP-betas estimates						
Intercept	0.26 [19.42]	0.26 [19.49]	0.26 [21.09]	0.26 [33.20]	0.26 [34.52]	0.26 [33.03]
$z_{\sigma_{TED}}$	-0.01 [-0.94]		0.02 [2.02]		0.00 [0.70]	
z_{TED}		-0.01 [-3.59]	-0.03 [-3.26]			-0.00 [-0.02]
$\ln(\sigma_1^{MKT}/\sigma_5^{MKT})$				-0.04 [-4.03]	-0.04 [-3.93]	-0.04 [-3.81]
adj.-R ² (%)	0.7	7.4	11.8	56.8	57.1	56.7

by sorting directly on the beta estimator (0.50 versus 0.67–0.68).

Appendix F. Alternative measures of FP-beta dispersion

Table F.12 investigates the power of different variables to predict FP-beta dispersion, measured using the cross-sectional standard deviation and mean absolute deviation in individual stocks' estimated FP-betas. It shows similar results to Panel A of Table 4.

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